PRIME POLYNOMIALS IN LEGENDRE INTERVALS OVER $\mathbb{F}_{2^K}[T]$, IV: POINT COUNTS ON HEXAD MODULAR CURVES, BROWN INVARIANTS OF WITT SHEAVES, AND THE CONSTANT X

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ABSTRACT. Let $q = 2^k$ with k odd. Part III reduced the closed form $X = q^6 + q^5$ of the second constant of the degree-6 Legendre variance to two statements about the sextic family $B_\lambda = \lambda_5^2 u^6 + \lambda_3^2 u^2 + \lambda_5 u + 1$ whose roots form the hexad of $\ker L'_\lambda$: the polynomial frequencies of its eleven factorisation types over $\mathbb{F}_q$, and the signs of the Gauss sums s_4, s_6 of explicit $\mathbb{Z}/4$-valued quadratic forms on $\mathbb{F}_{q^4}$, $\mathbb{F}_{q^6}$. This paper proves three structural theorems about these Gauss sums and uses them to settle the signs in the two classes of order 4. The *hexad formula* expresses s_{2m}, a sum over $\mathbb{F}_{q^{2m}}$, as q^m times a signed sum of at most sixteen fourth roots of unity indexed by the $\mathbb{F}_{q^m}$-rational points of the coset $T_\lambda'^{-1}(1)$ (the six roots and the ten synthematic totals), with sign $+$ at roots and $-$ at totals. The *trace parity theorem* determines the absolute trace of every such point: $\mathrm{Tr}_{q^m/2}(w) \equiv mk + \dim(\mathcal{K}_\lambda \cap \mathbb{F}_{q^m}) \pmod 2$; for $m = 1$ it says that a rational root of B_λ has trace 0 or 1 according to the Frobenius class. Together with an elementary count on the sixteen-point configuration of roots and totals they determine the signs in nine of the eleven classes: $s_4 = +4q^2$ throughout the classes 1^6, 2^3; $s_4 = \pm 4q^2$ on complementary half-spaces of λ_1 in the classes $2\,1^4$, $2^2 1^2$; $s_4 \in \{0, +4q^2\}$, respectively $\{0, -4q^2\}$, nonzero exactly on an explicit affine hyperplane, in the classes $4\,1^2$, $4\,2$, hence $W' = +2q^7$ and $-2q^7$ as conjectured since Part II; and $S'' = +4q^6$ throughout the classes $3\,1^3$ and 3^2. The class 6 is settled under a single Witt identity verified for $q \le 32$, and only the class $3\,2\,1$ remains open. The hyperplane parameter κ_0 of the order-4 classes is given in closed form in the ground-field data of the quartic cofactor; contrary to what the data at $q = 8$ suggested, it is not constant on the class. For the factorisation counts, every moment $\sum_\lambda N_m(B_\lambda)$ is expressed through Gauss sums of Gold-type forms on the trace-zero hyperplane of $\mathbb{F}_{q^m}$; the identity is proved for all m, the third moment $2q^2 - 3q - 1$ is verified for $q \le 128$, and an explicit nonsingular 11×11 system of pure and mixed moments whose unique solution is the conjectured table of class counts is written down.

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Date: October 11, 2026.

DOI 10.5281/zenodo.23299284. Part I: DOI 10.5281/zenodo.23225896; Part II: DOI 10.5281/zenodo.23252762; Part III: DOI 10.5281/zenodo.23290724. Code and data for this paper: https://github.com/Ruqing1963/legendre-intervals-part4.

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1. Introduction and main results

1.1. **Where Part III left the problem.** Parts I–III [7, 8, 9] study the variance of the prime-power count Ψ over Legendre intervals $\{f^2 + s : \deg s \le d\}$ in $\mathbb{F}_q[t]$, $q = 2^k$, the function-field analogue of the problem of Keating and Rudnick [18] in the even-characteristic regime where the Weil-II/equidistribution route of Katz [16, 17] must be replaced by exact evaluations. At $d = 6$ the variance is an explicit function of three constants A_6, X, Y (Part II); $A_6 = q^9 + (q-1)q^5$ is proved for odd k, $Y = q^3 m_k$ has no elementary closed form, and

$$X = q^{-3} \sum_{\lambda_3, \lambda_5 \in \mathbb{F}_q} W'(\lambda_3, \lambda_5), \qquad W'(\lambda_3, \lambda_5) = \sum_{\lambda_1 \in \mathbb{F}_q} S''(\lambda_1, \lambda_3, \lambda_5), \tag{1}$$

where $S''(\lambda) = \sum_{x \in \mathbb{F}_{q^{12}}} i^{\mathfrak{q}_\lambda(x)}$ is the Gauss sum of the $\mathbb{Z}/4$-valued quadratic form

$$\mathfrak{q}_\lambda(x) = \mathsf{T}_{12k}(x) + 2\,\mathrm{Tr}_{q^{12}/2}(\lambda_5 x^5 + \lambda_3 x^3 + \lambda_1 x), \qquad \mathsf{T}_n(x) = \mathrm{Tr}_{W_2(\mathbb{F}_{2^n})/\mathbb{Z}/4}[x],$$

$[x]$ the Teichmüller representative in the Witt vectors of length two (the trace of Frobenius on a rank-four Witt–Artin–Schreier sheaf in the sense of Brylinski [5] and Katz [16]; the quadratic forms with values in $\mathbb{Z}/4$ and their Gauss sums are those of Brown [4]). Part III proved that for $\lambda_5 \neq 0$ the radical $\mathcal{K}_\lambda$ of $\mathfrak{q}_\lambda$ is governed by the hexad $\{r_1, \dots, r_6\}$ of roots of

$$B_\lambda(u) = \lambda_5^2 u^6 + \lambda_3^2 u^2 + \lambda_5 u + 1, \qquad \mathcal{K}_\lambda = \ker T'_\lambda = \{0\} \cup \{r_i + r_j\}, \qquad T'_\lambda(r_i) = 1,$$

where $T'_\lambda(x) = \lambda_5 x^4 + \lambda_3 x^2 + x + (\lambda_3 x)^{1/2} + (\lambda_5 x)^{1/4}$ is the polar operator of $\mathfrak{q}_\lambda$, i.e.

$$\mathfrak{q}_\lambda^{(m)}(x+y) = \mathfrak{q}_\lambda^{(m)}(x) + \mathfrak{q}_\lambda^{(m)}(y) + 2\,\mathrm{Tr}_{q^m/2}\big(y\,T'_\lambda(x)\big) \qquad (x, y \in \mathbb{F}_{q^m}), \tag{2}$$

for the forms $\mathfrak{q}_\lambda^{(m)}(x) = \mathsf{T}_{mk}(x) + 2\,\mathrm{Tr}_{q^m/2}(\lambda_5 x^5 + \lambda_3 x^3 + \lambda_1 x)$ on $\mathbb{F}_{q^m}$ with Gauss sums $s_m(\lambda) = \sum_{x \in \mathbb{F}_{q^m}} i^{\mathfrak{q}_\lambda^{(m)}(x)}$. The Frobenius σ acts on $\mathcal{K}_\lambda$ through $S_6 \cong \mathrm{Sp}_4(\mathbb{F}_2)$ [10] as the permutation π_λ of the roots, the coset $T'^{-1}_\lambda(1) = r_1 + \mathcal{K}_\lambda$ consists of the six roots and the ten *synthematic totals* $r_i + r_j + r_l$ ($\{i,j\}, \{l, \cdot\}$ running over the three duads of a syntheme), and W' is determined by the class of π_λ together with the signs of certain small-field Gauss sums. Precisely (Part III, Theorems 6.9, 6.12, 6.14 and Corollary 1.5), with N_0 the order of π_λ:

- class 5 1: $W' = -q^7$ (proved);

- $N_0 \mid 4$ (classes 1^6, $2\,1^4$, $2^2 1^2$, 2^3, $4\,1^2$, $4\,2$): $S''(\lambda) = q^4 s_4(\lambda)$ with $s_4(\lambda) \in \{0, \pm 4q^2\}$ (proved); for $N_0 = 4$ the set $\{\lambda_1 : s_4 \neq 0\}$ is empty or the affine hyperplane $\mathrm{Tr}_{q/2}(\lambda_1 \tau_0) = \kappa_0$, $\tau_0 = r_5 + r_6$ (proved); the *signs* were conjectural;
- $N_0 \in \{3, 6\}$ (classes $3\,1^3$, 3^2, $3\,2\,1$, 6): $S''(\lambda) = \pm 4q^6$ for all λ_1 (proved); the sign is $+$ when $s_3 \neq 0$ and $N_0 = 3$ (proved), and is otherwise conjectural.

Target Theorem 1.1 (Part III, Conjecture 5.3). *For odd k the number of $(\lambda_3, \lambda_5) \in \mathbb{F}_q^\times \times \mathbb{F}_q^\times$ with B_λ of class C is the polynomial $n_C(q)$ of Part III, Table 2:*

$$\begin{array}{llllll} 1^6: \frac{(q-2)(q-8)}{720}, & 2\,1^4: \frac{(q-2)(q-8)}{48}, & 2^2 1^2: \frac{q(q-2)}{16}, & 2^3: \frac{(q-2)(q-8)}{48}, & & \\ 3\,1^3: \frac{(q+1)(q-2)}{18}, & 3^2: \frac{(q+1)(q-2)}{18}, & 3\,2\,1: \frac{(q+1)(q-2)}{6}, & 4\,1^2: \frac{(q-2)(q-4)}{8}, & & \\ 4\,2: \frac{q(q-2)}{8}, & 5\,1: \frac{q^2+1}{5}, & 6: \frac{(q+1)(q-2)}{6}, & & & \end{array}$$

and for $\lambda_3 = 0$ the classes 2^3, $4\,1^2$, 6 occur $(q-2)/6$, $(q-2)/2$, $(q+1)/3$ times (the latter is a theorem). Status: verified for $q \leq 512$; the relations of Part III, Corollary 5.1, and the moment identities of Section 2 are proved.

Target Theorem 1.2 (Part III, Conjecture 6.17). *For odd k: $s_4(\lambda) = +4q^2$ for all λ_1 in the classes 1^6, 2^3; $s_4(\lambda) = \pm 4q^2$ for $q/2$ values of λ_1 each in the classes $2\,1^4$, $2^2 1^2$; $s_4(\lambda) \in \{0, +4q^2\}$ in the class $4\,1^2$ and $\in \{0, -4q^2\}$ in the class $4\,2$, nonzero for $q/2$ values of λ_1; $S''(\lambda) = +4q^6$ for all λ_1 in the classes $3\,1^3$, 3^2, 6, and $= \pm 4q^6$ in equal numbers in the class $3\,2\,1$.* Status: proved in this paper for the classes 1^6, $2\,1^4$, $2^2 1^2$, 2^3 (Theorem 3.17), $4\,1^2$, $4\,2$ (Theorem 3.9), $3\,1^3$, 3^2 (Theorem 3.20), and for the class 6 under the hypothesis (H) of (9), verified at $q = 8, 32$; the class $3\,2\,1$ remains verified only (all λ at $q = 8$, and through W' for $q \leq 128$).

Corollary 1.3 (Part III, Corollary 1.5). *Target Theorems 1.1 and 1.2 imply $X = q^6 + q^5$ for all odd k, and hence*

$$\frac{\mathrm{Var}_{\mathcal{G}^2}(\Psi)}{q^7} = \frac{(q-1)^2(q+2)^2}{q^2} + \frac{(q-1)(4q^2 + m_k^2)}{q^4} \qquad (k \text{ odd}),$$

with $m_k = Y/q^3$ the only remaining unevaluated quantity.

1.2. **Results of this paper.** The paper has two halves, corresponding to the two targets. Everything stated as a theorem, proposition or lemma below is proved; numerical statements carry the word *numerical* in their header, with the range of verification.

Gauss sums (Section 3).

(i) **The hexad formula** (Theorem 3.3). For $\lambda_5 \neq 0$ and $m \geq 1$,

$$s_{2m}(\lambda) = q^m \sum_{w \in T_\lambda'^{-1}(1) \cap \mathbb{F}_{q^m}} \varepsilon(w)\, i^{\mathfrak{q}_\lambda^{(m)}(w)}, \qquad \varepsilon(w) = \begin{cases} +1, & B_\lambda(w) = 0, \\ -1, & \text{otherwise.} \end{cases}$$

The sum over q^{2m} elements is thus a signed sum of at most sixteen fourth roots of unity, attached to the rational points of the hexad coset in $\mathbb{F}_{q^m}$; the signs distinguish roots from synthematic totals. It is verified for every λ at $q = 8$ in all classes with $N_0 \mid 4$ ($m = 2$) and $N_0 \in \{3, 6\}$ ($m = 3$).

(ii) **Trace parity** (Theorem 3.5). If $w \in T_\lambda'^{-1}(1) \cap \mathbb{F}_{q^m}$ and $s_m(\lambda_1') \neq 0$ for some λ_1', then $\mathrm{Tr}_{q^m/2}(w) \equiv mk + f_m \pmod 2$, $f_m = \dim_{\mathbb{F}_2}(\mathcal{K}_\lambda \cap \mathbb{F}_{q^m})$. For $m = 1$ the hypothesis is automatic: every rational root of B_λ has absolute trace $f_1(C) + 1 \bmod 2$ (k odd), i.e. trace 0 in the classes $2\,1^4$, $3\,2\,1$, $4\,1^2$ and trace 1 in 1^6, $2^2 1^2$, $3\,1^3$, $5\,1$ (Corollary 3.6; verified for every (a, b) at $q = 8, 32, 128$). This refines the pair lemma of Part III.

(iii) **The classes of order** 4 (Theorem 3.9). For every λ_1,

$$s_4(\lambda_1) = 2q^2\big(i^{\mathfrak{q}_\lambda^{(2)}(r_5)} - i^{\mathfrak{q}_\lambda^{(2)}(w_3)}\big), \qquad w_3 = r_5 + r_1 + r_3,$$

and $\mathfrak{q}_\lambda^{(2)}(r_5) = 0$ in the class $4\,1^2$, $\mathfrak{q}_\lambda^{(2)}(w_3) = 0$ in the class $4\,2$. Hence $s_4 \in \{0, +4q^2\}$, resp. $\{0, -4q^2\}$, $\operatorname{sgn} s_4 = (-1)^{\mathrm{Tr}_{q/2}(\rho_0/\tau_0^2)}$ ($\rho_0 = r_5 r_6$) and $W' = +2q^7$, resp. $-2q^7$: the order-4 part of Target Theorem 1.2. The parameter κ_0 of the hyperplane $\{s_4 \neq 0\}$ is computed in closed form (Proposition 3.12),

$$\kappa_0 = \mathrm{Tr}_{q/2}\Big(z^3 + \frac{\rho_0^2}{z} + \frac{(z+1)\tau_0^2}{\rho_0}\Big), \qquad z = \frac{\tau_0}{\lambda_5\rho_0}, \quad z^2 + z = \tau_0^4,$$

and takes both values on each class as soon as $q \geq 32$ (Table 3); the statement $\kappa_0 = 0$ suggested by the data at $q = 8$ is false.

(iv) **The other classes** (Theorems 3.17, 3.20). On the sixteen-point coset the values $\phi(w) = \frac{1}{2}(\mathfrak{q}^{(m)}(w) - p)$ are affine-linear in the duads (Lemma 3.16), so the signed sum of the hexad formula is ± 4 according to the number n_1 of roots with $\phi = 1$ modulo 4; trace parity pins ϕ at the rational roots and totals. This gives $s_4 = +4q^2$ for all λ_1 in the classes 1^6, 2^3, the half-half law in $2\,1^4$, $2^2 1^2$, and $S'' = +4q^6$ for all λ_1 in $3\,1^3$, 3^2; for the class 6 the same holds under one Witt identity (H) verified for $q \leq 32$. The class $3\,2\,1$ is the only one left.

Counts (Section 2).

(v) Proposition 2.6: for every m, $\sum_\lambda N_m(B_\lambda) = (q-1)^2 - q[2 \mid m] + q^{2-m}\sum_{\theta\in\Theta_m} G_{H_m}(\theta)$, where $G_{H_m}(\theta)$ is the Gauss sum of the Gold-type form $e \mapsto \mathrm{Tr}(\theta e^5 + \theta e^3)$ on the trace-zero hyperplane of $\mathbb{F}_{q^m}$ and Θ_m is a trace condition on the Gold parameter $1/\theta$; the cases $m = 1, 2$ recover Part III, and $m = 3$ gives $2q^2 - 3q - 1$ (verified $q \leq 128$, Conjecture 2.4, with the structure of the sums in Proposition 2.2).

(vi) Proposition 2.8: the moments $m \leq 6$ form a linear system of rank 6 in the eleven class counts, and the Gauss-sum totals predicted by Target Theorem 1.1 are listed; the proof of the target is thereby reduced to six explicit Gauss-sum evaluations plus the mixed moments.

Section 4 records what the two targets now amount to, and Section 5 discusses the interpretation of the hexad as the 2-torsion of a lift of the supersingular Jacobian, including the mechanism behind the dependence of the even-k counts on k mod 4.

1.3. **Notation.** $q = 2^k$, k odd unless said otherwise; $\sigma(x) = x^q$; $\mathrm{Tr} = \mathrm{Tr}_{q^m/2}$ when m is clear; $\tau_m = \mathrm{Tr}_{q^m/q}$. For a permutation class C of S_6, $f_m(C)$ is the number of fixed points of π^m on $\{1, \dots, 6\}$ and, by abuse, also $\dim_{\mathbb{F}_2} \mathcal{K}_\lambda^{\sigma^m} = \log_2(1 + \#\{\text{duads fixed by } \pi^m\})$; the two are listed in Table 1. T_n is the Witt trace, $\mathsf{T}_n(x) = \mathrm{Tr}_{2^n/2}(x) + 2e_2^{\mathrm{abs}}(x)$ with $e_2^{\mathrm{abs}}(x) = \sum_{i<j} x^{2^i+2^j}$, so that $\mathfrak{q}_\lambda^{(m)}(x) \equiv \mathrm{Tr}_{q^m/2}(x) \pmod 2$. All numerical statements refer to the scripts listed in the final section.

2. Higher moments and moduli of sextic factorisations

2.1. **The geometry of the family.** Write $F_{a,b}(u) = u^6 + au^2 + bu + b^2$, $(a, b) \in \mathbb{F}_q \times \mathbb{F}_q^\times$, for the normalised family ($a = \lambda_3^2/\lambda_5^2$, $b = \lambda_5^{-1}$), and $N_m(a, b) = \#\{u \in \mathbb{F}_{q^m} : F_{a,b}(u) = 0\}$. For $u \neq 0$ the equation $F_{a,b}(u) = 0$ is linear in a:

$$a = g_b(u) := u^4 + \frac{b}{u} + \frac{b^2}{u^2}, \tag{3}$$

so $\sum_{a,b} N_m(a, b) = \#\{(u, b) \in \mathbb{F}_{q^m}^\times \times \mathbb{F}_q^\times : g_b(u) \in \mathbb{F}_q\}$: the m-th moment counts the $\mathbb{F}_{q^m}$-points u at which the rational function g_b takes values in the ground field, i.e. the

TABLE 1. The eleven classes of π_λ: order N_0, fixed points $f^{(m)}$ of π^m on the roots, and $f_m = \dim \mathcal{K}_\lambda^{\sigma^m}$ for $m = 1, 2, 3$.

class	1^6	$2\,1^4$	$2^2 1^2$	2^3	$3\,1^3$	3^2	$3\,2\,1$	$4\,1^2$	$4\,2$	$5\,1$	6
N_0	1	2	2	2	3	3	6	4	4	5	6
$f^{(1)}$	6	4	2	0	3	0	1	2	0	1	0
$f^{(2)}$	6	6	6	6	3	0	3	2	2	1	0
$f^{(3)}$	6	4	2	0	6	6	4	2	0	1	0
f_1	4	3	2	2	2	0	1	1	1	0	0
f_2	4	4	4	4	2	0	1	2	2	0	0
f_3	4	3	2	2	4	4	3	1	1	0	2

$\mathbb{F}_q$-points of the curve $C_m^{(b)} = \{(u, a) : a = g_b(u)\} \subset \mathrm{Res}_{\mathbb{F}_{q^m}/\mathbb{F}_q}\mathbb{A}^1 \times \mathbb{A}^1$. The class counts n_C are determined by the joint distribution of the fixed-point numbers $(f_C^{(1)}, \dots, f_C^{(6)})$, and the mixed moments $\sum_\lambda \prod_m N_m^{e_m}$ are counts of tuples of roots, i.e. point counts on fibre products of the $C_m^{(b)}$ over the a-line; these fibre products are the "hexad modular curves" of the title. Figure 1 summarises how the moments are evaluated in this section.

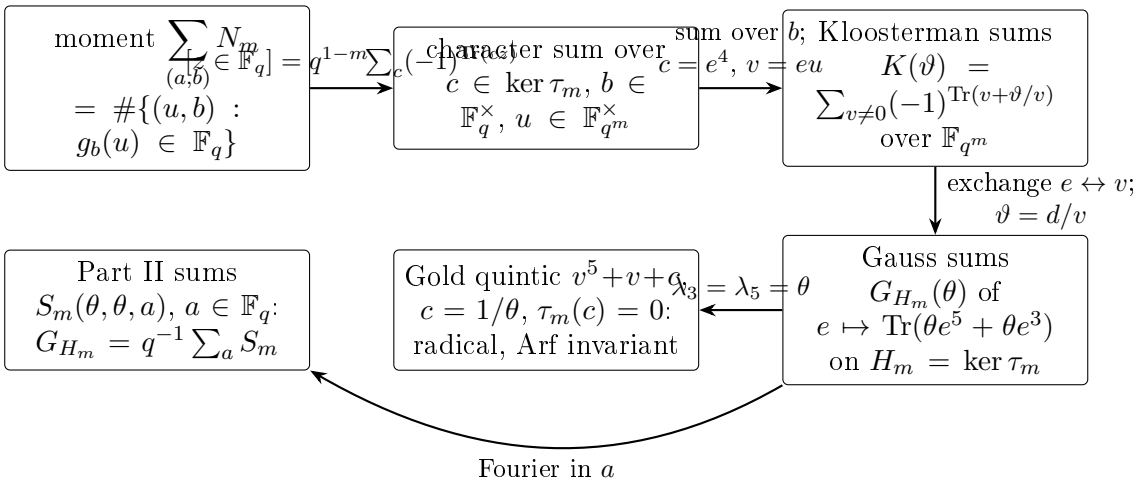


FIGURE 1. The evaluation of the moments $\sum N_m$ (Propositions 2.1, 2.6): every step is an exact identity; the only quantities that are not elementary are the Gauss sums $G_{H_m}(\theta)$, which are 0 or $\pm 2^{(mk+r)/2}$ with r the dimension of the radical on the hyperplane. No archimedean estimate enters anywhere, which is why the moments can be exact polynomials in q.

2.2. **The third moment.** Let $\mathrm{Tr} = \mathrm{Tr}_{q^3/2}$, $\tau = \mathrm{Tr}_{q^3/q}$, $H = \ker\tau \subset \mathbb{F}_{q^3}$ (an $\mathbb{F}_2$-space of dimension $2k$), and for $\theta \in \mathbb{F}_{q^3}$ put

$$G_H(\theta) = \sum_{e\in H}(-1)^{\mathrm{Tr}(\theta e^5+\theta e^3)}, \qquad \Theta = \{\theta \in \mathbb{F}_{q^3}^\times : \tau(\theta^{-1}) = 0\}.$$

$G_H(\theta)$ is the Gauss sum of the $\mathbb{F}_2$-quadratic form $Q_\theta(e) = \mathrm{Tr}(\theta e^5 + \theta e^3)$ (both exponents are of Gold type 2^i+1; the polynomials $v^{2^i+1}+v+c$ are those of Bluher [3] and Helleseth–Kholosha [14]) restricted to the hyperplane H; equivalently $G_H(\theta) = q^{-1}\sum_{a\in\mathbb{F}_q} S_3(\theta, \theta, a)$ with $S_3(\lambda_5, \lambda_3, \lambda_1) = \sum_{e\in\mathbb{F}_{q^3}}(-1)^{\mathrm{Tr}(\lambda_5 e^5+\lambda_3 e^3+\lambda_1 e)}$ the sums of Part II over $\mathbb{F}_{q^3}$ (the curves $y^2 + y = \lambda_5 x^5 + \lambda_3 x^3$ are supersingular, cf. van der Geer–van der Vlugt [12]).

Proposition 2.1. *For every $q = 2^k$,*

$$\sum_{(a,b)\in\mathbb{F}_q\times\mathbb{F}_q^\times} N_3(a,b) = (q-1)^2 + \frac{1}{q}\sum_{\theta\in\Theta} G_H(\theta).$$

Proof. By (3), $\sum N_3 = \#\{(u,b)\in\mathbb{F}_{q^3}^\times\times\mathbb{F}_q^\times : g_b(u)\in\mathbb{F}_q\}$. The annihilator of $\mathbb{F}_q$ for the pairing $(c,z)\mapsto \mathrm{Tr}(cz)$ on $\mathbb{F}_{q^3}$ is $A=\{c:\tau(c)=0\}$, of order q^2, so $[z\in\mathbb{F}_q] = q^{-2}\sum_{c\in A}(-1)^{\mathrm{Tr}(cz)}$ and

$$\sum N_3 = \frac{1}{q^2}\sum_{c\in A}\sum_{b\in\mathbb{F}_q^\times}\sum_{u\neq 0}(-1)^{\mathrm{Tr}(cu^4+cb/u+cb^2/u^2)} = \frac{1}{q^2}\sum_{c\in A}\sum_{b}\sum_{u}(-1)^{\mathrm{Tr}(c^{1/4}u)+\mathrm{Tr}(b(c+c^{1/2})/u)},$$

using $\mathrm{Tr}(z^4)=\mathrm{Tr}(z^2)=\mathrm{Tr}(z)$. The sum over $b\in\mathbb{F}_q^\times$ is $q[\tau((c+c^{1/2})/u)=0]-1$. The part with -1 contributes $-q^{-2}\sum_{c\in A}\sum_{u\neq0}(-1)^{\mathrm{Tr}(c^{1/4}u)} = -q^{-2}\big((q^3-1)-(q^2-1)\big) = -(q-1)$. In the other part the term $c=0$ contributes $(q^3-1)/q$; for $c\neq0$ write $c=e^4$ with $e\in A\backslash 0$ and substitute $v=eu$: the condition becomes $\tau((e^5+e^3)/v)=0$ and the weight $(-1)^{\mathrm{Tr}\,v}$, so this part equals

$$\frac{q^3-1}{q} + \frac{1}{q^2}\sum_{e\in A\backslash 0}\sum_{d\in\mathbb{F}_q} K\big(d(e^5+e^3)\big), \qquad K(\vartheta) = \sum_{v\neq0}(-1)^{\mathrm{Tr}(v+\vartheta/v)}$$

(Kloosterman sums over $\mathbb{F}_{q^3}$, cf. Carlitz [6] and Lachaud–Wolfmann [19] for their arithmetic in characteristic 2). The terms $d=0$ give $-(q^2-1)/q^2$. For $d\neq0$ we exchange the order: $\sum_{e\in A\backslash0}(-1)^{\mathrm{Tr}(d(e^5+e^3)/v)} = G_H(d/v)-1$, and with $\vartheta=d/v$,

$$\sum_{d\neq0}\sum_{v\neq0}(-1)^{\mathrm{Tr}\,v}\big(G_H(d/v)-1\big) = (q-1) + \sum_{\vartheta\neq0} G_H(\vartheta)\big(q[\tau(\vartheta^{-1})=0]-1\big).$$

Finally $\sum_{\vartheta\neq0}G_H(\vartheta) = q^3\#\{e\in H: e^5+e^3=0\} - |H| = q^3-q^2$, since $e^5+e^3=e^3(e+1)^2$ vanishes only at $e\in\{0,1\}$ and $1\notin H$ ($\tau(1)=3=1$). Collecting,

$$\begin{aligned}\sum N_3 &= -(q-1) + \frac{q^3-1}{q} - \frac{q^2-1}{q^2} + \frac{(q-1)-(q^3-q^2)}{q^2} + \frac{1}{q}\sum_{\theta\in\Theta} G_H(\theta)\\ &= (q-1)^2 + \frac{1}{q}\sum_{\theta\in\Theta} G_H(\theta).\end{aligned}$$

□

Proposition 2.2 (structure of $G_H(\theta)$; numerical at $q=8,32,128$)**.** *Let $\theta\in\Theta$ and $c=\theta^{-1}$ (so $\tau(c)=0$), and let (`moment3_structure.py`)*

$$R = \ker T_\theta \cap \mathbb{F}_{q^3}, \qquad R_H = \{e\in H : T_\theta(e)\in\mathbb{F}_q\}$$

be the radical of Q_θ on $\mathbb{F}_{q^3}$ (the σ^3-fixed part of the quintic frame of v^5+v+c, of dimension 1 for the Gold types $(1,4)$, $(2,3)$ and 3 for $(1,1,1,2)$) and the radical of the restriction $Q_\theta|_H$ (the orthogonal of H for the absolute trace pairing is $\mathbb{F}_q$). For all $\theta\in\Theta$ at $q=8$, 32 and 128:

(i) $\dim R_H = 2$, *whatever the Gold type and* $\dim R$;
(ii) $G_H(\theta)\neq0$ *iff Q_θ vanishes on R_H;*
(iii) *whenever $G_H(\theta)\neq0$ the induced nondegenerate form on H/R_H ($\dim 2k-2$) has Arf invariant 0 [1], so $G_H(\theta) = +2^{k+1}$;*

(iv) *the joint distribution of the Gold type of* v^5+v+c *over* $\mathbb{F}_{q^3}$ *and of the vanishing of* $Q_\theta|_{R_H}$ *is*

	$(1,4)$		$(2,3)$		$(1,1,1,2)$	
	$G_H\neq 0$	$G_H=0$	$G_H\neq 0$	$G_H=0$	$G_H\neq 0$	$G_H=0$
$q=8$	27	27	0	9	0	0
$q=32$	285	255	90	243	120	30
$q=128$	4347	4095	2436	2901	1344	1260

with totals 27, 495, 8127 *of nonzero values, i.e.* $(q+1)(q-2)/2$ *in each case.*

Remark 2.3. The per-type counts $(54,9,0)$, $(540,333,150)$, $(2604,\dots)$ are not the values of a common quadratic polynomial with small denominators (the interpolating quadratics have coefficients like 331/640; `part4_moment_system.py`), whereas the total number of θ with $G_H(\theta)\neq 0$ is the clean polynomial $(q+1)(q-2)/2$. A proof of Conjecture 2.4 must therefore count the vanishing locus of $Q_\theta|_{R_H}$ directly rather than type by type.

Conjecture 2.4 (verified for $q=8,32,128$; `moment3_check.py`)**.** *For odd* k *the four statements of Proposition 2.2 hold for every* q*; in particular* $G_H(\theta)\in\{0,2^{k+1}\}$ *and* $\#\{\theta\in\Theta : G_H(\theta)\neq 0\}=(q+1)(q-2)/2$*. Consequently*

$$\sum_{(a,b)} N_3(a,b) = (q-1)^2+(q+1)(q-2) = 2q^2-3q-1,$$

which is the value predicted by Target Theorem 1.1 (Proposition 2.8). At $q=8$*:* $103=49+54$*; at* $q=32$*:* $1951=961+990$*; at* $q=128$*:* $32383=16129+16254$*.*

Remark 2.5 (what a proof requires)**.** Because $\theta\in\mathbb{F}_{q^3}\setminus\mathbb{F}_q$, the operator T_θ does *not* commute with σ, and the description of $\{e : T_\theta(e)\in\mathbb{F}_q\}$ through $(\sigma-1)^{-1}(\ker T_\theta)$ that drives Part II (where the parameters lie in $\mathbb{F}_q$) is not available; this is why $\dim R_H=2$ although $\dim R=1$. Three facts are needed: (a) a description of the two-dimensional radical R_H, uniform in the Gold type; (b) the condition "$Q_\theta|_{R_H}=0$" as an explicit condition on c and the count $(q+1)(q-2)/2$ of the $c\in\ker\tau$ satisfying it; (c) the vanishing of the Arf invariant. Through $G_H(\theta)=q^{-1}\sum_a S_3(\theta,\theta,a)$, (c) is a sign pattern of the Part II sums along $a\in\mathbb{F}_q$, which the shift calculus of Part II (with $c\in\mathbb{F}_{q^3}$, $T_\theta(c)\in\mathbb{F}_q$) should determine as in Part II, Theorem 6.12. With $(\lambda_3,\lambda_5)=(\theta,\theta)$ the Bluher quintic of Part II is $\theta(\theta u^5+\theta u+1)$, i.e. the Gold quintic v^5+v+c with $c=\theta^{-1}$ and no scaling, so the condition $\theta\in\Theta$ is exactly the trace condition $\tau(c)=0$ on the Gold parameter; Part II, Proposition 6.8 answers the analogous question over $\mathbb{F}_q$ (there $(q-2)/2$ Gold parameters are of type $(1,4)$), and the present question concerns the quintic over the cubic extension with a trace constraint.

2.3. **All moments.**

Proposition 2.6. *For every* $m\geq 1$*, with* $\tau_m=\mathrm{Tr}_{q^m/q}$*,* $H_m=\ker\tau_m$*,* $\Theta_m=\{\theta\in\mathbb{F}_{q^m}^\times : \tau_m(\theta^{-1})=0\}$ *and* $G_{H_m}(\theta)=\sum_{e\in H_m}(-1)^{\mathrm{Tr}_{q^m/2}(\theta e^5+\theta e^3)}$*,*

$$\sum_{(a,b)} N_m(a,b) = (q-1)^2 - q\,[2\mid m] + q^{2-m}\sum_{\theta\in\Theta_m} G_{H_m}(\theta).$$

Proof. The proof of Proposition 2.1 goes through with 3 replaced by m: $|A|=q^{m-1}$, the part with -1 gives $-(q-1)$, the term $c=0$ gives $q^{2-m}(q^m-1)$, the terms $d=0$ give $-q^{1-m}(q^{m-1}-1)$, the terms $d\neq 0$ give $q^{1-m}\big[(q-1)+q\sum_{\Theta_m}G_{H_m}-\sum_{\vartheta\neq 0}G_{H_m}(\vartheta)\big]$ with $\sum_{\vartheta\neq 0}G_{H_m}(\vartheta)=q^m\#\{e\in H_m : e^5+e^3=0\}-q^{m-1}$, and $\#\{e\in H_m : e^3(e+1)^2=0\}=1+[2\mid m]$ because $1\in H_m$ iff $\tau_m(1)=m$ vanishes in $\mathbb{F}_2$. Collecting the constants gives $(q-1)^2-q[2\mid m]$. □

Remark 2.7 (consistency with Part III, Theorem 1.3)**.** For $m=1$: $\tau_1=\mathrm{id}$, so $\Theta_1=\varnothing$ and $\sum N_1=(q-1)^2$. For $m=2$: $H_2=\mathbb{F}_q$ and $\Theta_2=\mathbb{F}_q^\times$; for $\theta,e\in\mathbb{F}_q$ one has $\mathrm{Tr}_{q^2/2}(\theta(e^5+e^3))=\mathrm{Tr}_{q/2}(2\theta(e^5+e^3))=0$, hence $G_{H_2}(\theta)=q$ for every $\theta\in\Theta_2$ and

$$\sum N_2=(q-1)^2-q+(q-1)q=2q^2-4q+1=(q-1)^2+q(q-2),$$

which is $\sum N_1+\sum(N_2-N_1)$ with the values of Part III, Theorem 1.3. Thus the two moments proved in Part III by root counting are the cases in which the Gauss sums degenerate ($H_1=0$; $H_2=\mathbb{F}_q$ with the form identically zero), and $m=3$ is the first case in which the supersingular sums of Part II genuinely enter.

2.4. **The elimination system.** Let $n_C(q)$ be the class counts (mixed and pure pairs together) and $f_C^{(m)}$ the fixed-point numbers of Table 1. Since $N_m(B_\lambda)=f_{C(\lambda)}^{(m)}$,

(4)
$$\sum_{(a,b)}N_m=\sum_C n_C(q)f_C^{(m)},\qquad \Sigma_m(q):=\sum_{\theta\in\Theta_m}G_{H_m}(\theta)=q^{m-2}\Big(\sum_{(a,b)}N_m-(q-1)^2+q[2\mid m]\Big).$$

Proposition 2.8 (`part4_moment_system.py`)**.** (i) *The 6×11 matrix $(f_C^{(m)})_{1\le m\le6}$ has rank 6; the six moments $m\le6$ determine the eleven class counts up to a five-dimensional space. Adjoining the five mixed moments $\sum N_1N_2$, $\sum N_1N_3$, $\sum N_2^2$, $\sum N_1^2N_2$ and $\sum N_1N_2N_3$ raises the rank to 10 only, whereas adjoining the five statistics*

$$M_{3,3}=\sum N_3^2,\qquad M_{2,3}=\sum N_2N_3,\qquad M_{2,2}=\sum N_2^2,$$
$$M_{1,3}=\sum N_1N_3,\qquad M_{3,3,3}=\sum N_3^3$$

gives an 11×11 matrix of determinant $1433272320=2^{17}\cdot3^7\cdot5\neq0$. Hence the eleven class counts are the unique solution of the linear system formed by the six pure moments and these five mixed moments, which are point counts on fibre products of the curves $C_m^{(b)}$ over the a-line (the two-fold products $C_3\times_{\mathbb{A}^1}C_3$, $C_2\times_{\mathbb{A}^1}C_3$, $C_2\times_{\mathbb{A}^1}C_2$, $C_1\times_{\mathbb{A}^1}C_3$ and the triple product of C_3). If the counts are those of Target Theorem 1.1, the five mixed moments are

$$M_{3,3}=8q^2-12q-7,\qquad M_{2,3}=5q^2-11q+3,\qquad M_{2,2}=7q^2-18q+9,$$
$$M_{1,3}=3q^2-7q+3,\qquad M_{3,3,3}=38q^2-54q-43,$$

and back-substitution of these eleven values returns the table. Target Theorem 1.1 is therefore equivalent to eleven explicit point counts: six on the curves $C_m^{(b)}$ and five on their fibre products.

(ii) *If the counts are those of Target Theorem 1.1, then the moments and the Gauss-sum totals are*

m	$\sum_{(a,b)}N_m$	$\Sigma_m=\sum_{\Theta_m}G_{H_m}$
1	$(q-1)^2$	0
2	$2q^2-4q+1$	$q(q-1)$
3	$2q^2-3q-1$	$q(q+1)(q-2)$
4	$3q^2-6q+1$	$q^3(2q-3)$
5	$2q^2-2q+2$	$q^3(q^2+1)$
6	$4q^2-4q-1$	$q^4(q-1)(3q+2)$

Conversely, Target Theorem 1.1 is equivalent to these six evaluations together with the five mixed moments; the rows $m=1,2$ are theorems (Part III), the row $m=3$ is Conjecture 2.4.

Proof. (i) is a rank computation (the matrix is printed by the script). (ii) is the substitution of the table into (4); the second column is then a direct consequence of Proposition 2.6. For the converse note that the class counts are integers determined by the eleven joint fixed-point statistics, which are linear combinations of the pure and mixed moments. □

Remark 2.9. The rows are a list of exact Gauss-sum identities to be proved. The row $m = 4$ reads $\sum_{\theta\in\mathbb{F}_{q^4}^\times,\ \tau_4(1/\theta)=0} G_{H_4}(\theta) = q^3(2q-3)$ with H_4 of dimension $3k$: the forms $e \mapsto \mathrm{Tr}_{q^4/2}(\theta e^5 + \theta e^3)$ have radicals controlled by the Gold quintic over $\mathbb{F}_{q^4}$, whose factorisation types over $\mathbb{F}_{q^4}$ are those of Part II, Proposition 6.8, extended by the splitting of $\mathbb{F}_{q^4}(\mu_5)$; the rows $m = 5, 6$ involve the Gold quintic over $\mathbb{F}_{q^5}$ and $\mathbb{F}_{q^6}$, where $\mu_5 \subset \mathbb{F}_{q^4}$ enters for $m \equiv 0 \bmod 4$ only. This is the precise sense in which the counts of Target Theorem 1.1 are polynomial: each Σ_m is a sum of terms $0, \pm 2^{(mk+r)/2}$ whose number is a polynomial point count on the Gold-quintic curves over $\mathbb{F}_{q^m}$ with a trace constraint, and the Arf invariants are constant on the strata; there is no cuspidal cohomology to control because every sum involved is supersingular (all normalised Frobenius eigenvalues are roots of unity), which is the structural reason for the absence of $O(q^{1/2})$ error terms. Turning this into a proof means establishing (a)–(c) of the remark after Conjecture 2.4 for each $m \le 6$ and for the fibre products; the authors of [16] reach polynomial answers in the odd-characteristic setting by Deligne's theory [11], which here is replaced by the exact Witt-vector evaluation.

3. Explicit Brown invariants of length-two Witt sheaves

3.1. **The shift calculus and representative cosets.** Recall from Part III (Lemma 6.4): for $c \in \mathbb{F}_{q^m}$ with $T'_\lambda(c) = a \in \mathbb{F}_q$,

$$s_m(\lambda_1) = i^{\mathfrak{q}_\lambda^{(m)}(c)}\, s_m(\lambda_1 + a) \qquad (\lambda_3, \lambda_5 \text{ fixed}), \tag{5}$$

which follows from (2) by substituting $x \mapsto x + c$. More generally, for any $c \in \mathbb{F}_{q^m}$ the same substitution gives $s_m(\lambda_1) = i^{\mathfrak{q}_\lambda^{(m)}(c)}\, s_m(\lambda_1 + T'_\lambda(c))$ with $\lambda_1 + T'_\lambda(c) \in \mathbb{F}_{q^m}$, where s_m is extended to parameters $\lambda_1 \in \mathbb{F}_{q^m}$ by the same formula.

Lemma 3.1. *Let $V_\lambda^{(m)} = \{c \in \mathbb{F}_{q^m} : T'_\lambda(c) \in \mathbb{F}_q\}$ and let $a_1, \dots, a_t$ be representatives of the cosets of $T'_\lambda(V_\lambda^{(m)})$ in $\mathbb{F}_q$. Then $\lambda_1 \mapsto s_m(\lambda_1)$ is determined by its values at $a_1, \dots, a_t$ together with the phases $\mathfrak{q}_\lambda^{(m)}(c)$, $c \in V_\lambda^{(m)}$; one element c per coset of $\mathcal{K}_\lambda \cap \mathbb{F}_{q^m}$ in $V_\lambda^{(m)}$ suffices. In particular, for $m = 4$ and $N_0 \mid 4$: $T'_\lambda(V_\lambda^{(4)}) = \mathbb{F}_q$ unless $N_0 = 4$, where it is the hyperplane $\tau_0^\perp$; a single value and the phases decide all of $\lambda_1 \mapsto s_4(\lambda_1)$.*

Proof. Immediate from (5); the image statements are Part III, Proposition 6.15(i). □

Two general facts about the Gauss sums will be used repeatedly. Let $R_m = \mathcal{K}_\lambda \cap \mathbb{F}_{q^m} = \mathcal{K}_\lambda^{\sigma^m}$, the radical of the polar form of $\mathfrak{q}_\lambda^{(m)}$ on $\mathbb{F}_{q^m}$ (by (2) and the nondegeneracy of the trace pairing), of dimension f_m.

Lemma 3.2. (i) *$\mathfrak{q}_\lambda^{(m)}$ is additive on R_m, and $s_m(\lambda) \neq 0$ iff $\mathfrak{q}_\lambda^{(m)}$ vanishes on R_m; in that case $|s_m(\lambda)|^2 = 2^{mk+f_m}$.*

(ii) *$\mathfrak{q}_{\lambda_1+1}^{(m)}(x) = \mathfrak{q}_{\lambda_1}^{(m)}(x) + 2\,\mathrm{Tr}_{q^m/2}(x)$ and $\mathfrak{q}_\lambda^{(m)}(x) \equiv \mathrm{Tr}_{q^m/2}(x) \pmod 2$; consequently $s_m(\lambda_1 + 1) = \overline{s_m(\lambda_1)}$.*

(iii) *Write $s_m(\lambda) = \mathrm{Re} + i\,\mathrm{Im}$ with*

$$\mathrm{Re} = \sum_{\mathrm{Tr}\,x=0} (-1)^{\mathfrak{q}(x)/2}, \qquad \mathrm{Im} = \sum_{\mathrm{Tr}\,x=1} (-1)^{(\mathfrak{q}(x)-1)/2}.$$

These are Gauss sums of $\mathbb{F}_2$-valued quadratic forms on the hyperplane $\ker \mathrm{Tr}_{q^m/2}$ *(the second one shifted by a constant), hence each lies in* $\{0\} \cup \{\pm 2^j : j \geq 0\}$ [21].

(iv) *There is* $\lambda_1' \in \mathbb{F}_q$ *with* $s_m(\lambda_1') \neq 0$ *if and only if* $\frac{1}{2}\mathfrak{q}^{(m)}_{\lambda,\lambda_1=0}$ *vanishes on* $R_m \cap \ker \tau_m$. *This holds for* $m = 1$.

Proof. (i) Additivity on R_m is (2). If $\mathfrak{q}(\kappa) \neq 0$ for some $\kappa \in R_m$, the substitution $x \mapsto x + \kappa$ multiplies s_m by $i^{\mathfrak{q}(\kappa)} \neq 1$, so $s_m = 0$. Otherwise $|s_m|^2 = \sum_{x,z} i^{\mathfrak{q}(x+z)-\mathfrak{q}(x)} = \sum_z i^{\mathfrak{q}(z)} \sum_x (-1)^{\mathrm{Tr}(xT'(z))} = q^m \sum_{z\in R_m} i^{\mathfrak{q}(z)} = q^m |R_m|$. (ii) The first identity is the definition, the congruence is $\mathsf{T}_{mk} \equiv \mathrm{Tr} \bmod 2$; the terms with $\mathrm{Tr}\, x = 0$ are real and unchanged under $\lambda_1 \mapsto \lambda_1 + 1$, those with $\mathrm{Tr}\, x = 1$ are purely imaginary and change sign. (iii) On $H_0 = \ker \mathrm{Tr}$ the values of $\mathfrak{q}$ are even and $Q_0 = \mathfrak{q}/2$ satisfies $Q_0(x+y) = Q_0(x) + Q_0(y) + \mathrm{Tr}(yT'(x))$ by (2), a quadratic form; on $x_1 + H_0$ one has $(\mathfrak{q}(x_1+h) - 1)/2 = (\mathfrak{q}(x_1)-1)/2 + Q_0(h) + \mathrm{Tr}(hT'(x_1))$. Gauss sums of $\mathbb{F}_2$-quadratic forms are 0 or $\pm 2^{(\dim + \mathrm{rad})/2}$. (iv) By (i), $s_m(\lambda_1) \neq 0$ iff $\mathfrak{q}_0(\kappa) + 2\,\mathrm{Tr}_{q/2}(\lambda_1 \tau_m(\kappa)) = 0$ for all $\kappa \in R_m$, i.e. iff the linear functional $\frac{1}{2}\mathfrak{q}_0$ on R_m equals $\kappa \mapsto \mathrm{Tr}_{q/2}(\lambda_1 \tau_m(\kappa))$; such a λ_1 exists iff $\frac{1}{2}\mathfrak{q}_0$ factors through $\tau_m|_{R_m}$, every linear functional on the subspace $\tau_m(R_m) \subset \mathbb{F}_q$ being of the form $\mathrm{Tr}_{q/2}(\lambda_1 \cdot)$. For $m = 1$, τ_1 is the identity. □

3.2. **The hexad formula.**

Theorem 3.3 (hexad formula)**.** *Let* $\lambda_5 \neq 0$, $m \geq 1$, *and let* $\varepsilon(w) = +1$ *if* $B_\lambda(w) = 0$ *and* $\varepsilon(w) = -1$ *otherwise. Then*

$$s_{2m}(\lambda) = q^m \sum_{w \in T_\lambda'^{-1}(1) \cap \mathbb{F}_{q^m}} \varepsilon(w)\, i^{\mathfrak{q}^{(m)}_\lambda(w)}. \tag{6}$$

Here $T_\lambda'^{-1}(1) = r_1 + \mathcal{K}_\lambda$ *consists of the six roots of* B_λ *and the ten synthematic totals, so the sum has at most sixteen terms; the set* $T_\lambda'^{-1}(1) \cap \mathbb{F}_{q^m}$ *is* $r + \mathcal{K}_\lambda^{\sigma^m}$ *for any* $r \in T_\lambda'^{-1}(1) \cap \mathbb{F}_{q^m}$, *and is empty if there is no such* r.

Proof. Write $V = \mathbb{F}_{q^{2m}}$, $U = \mathbb{F}_{q^m}$, $\mathfrak{q} = \mathfrak{q}^{(2m)}_\lambda$, $\mathrm{Tr} = \mathrm{Tr}_{q^{2m}/2}$, $\varphi = \sigma^m$ (so $\varphi(x) = x^{q^m}$ and $\mathrm{Tr}_{q^{2m}/q^m} = \varphi + 1$).

Step 1: $\mathfrak{q}$ on U. For $u \in U$, $\mathsf{T}_{2mk}(u) = \mathrm{Tr}_{W_2(U)/\mathbb{Z}/4}([u] + [u]) = \mathrm{Tr}_{W_2(U)/\mathbb{Z}/4}(0, u^2) = 2\,\mathrm{Tr}_{q^m/2}(u)$ and $2\,\mathrm{Tr}(f(u)) = 4\,\mathrm{Tr}_{q^m/2}(f(u)) = 0$; so $\mathfrak{q}(u) = 2\,\mathrm{Tr}_{q^m/2}(u)$.

Step 2: summation over the cosets of U. For $c \in V$, $u \in U$, by (2) and Step 1,

$$\mathfrak{q}(c+u) = \mathfrak{q}(c) + 2\,\mathrm{Tr}_{q^m/2}(u) + 2\,\mathrm{Tr}\big(uT'(c)\big) = \mathfrak{q}(c) + 2\,\mathrm{Tr}_{q^m/2}\big(u\,(1 + (\varphi+1)T'(c))\big),$$

hence $\sum_{u\in U} i^{\mathfrak{q}(c+u)} = i^{\mathfrak{q}(c)}\, q^m\, [(\varphi+1)T'(c) = 1]$, and $\mathfrak{q}$ is constant on the cosets $c+U$ with $(\varphi+1)T'(c) = 1$ (this condition depends only on the coset, since $T'(U) \subset U = \ker(\varphi+1)$). Therefore

$$s_{2m}(\lambda) = \sum_{c\in V/U} \sum_{u\in U} i^{\mathfrak{q}(c+u)} = q^m \sum_{\substack{c\in V/U \\ (\varphi+1)T'(c)=1}} i^{\mathfrak{q}(c)}.$$

Since T' has coefficients in $\mathbb{F}_q$ it commutes with φ, so $(\varphi+1)T'(c) = T'((\varphi+1)c)$, and the condition reads $w := c + c^{q^m} \in T'^{-1}(1)$. As $\varphi + 1$ maps V onto U with kernel U, the cosets $c + U$ correspond bijectively to the elements $w \in U$, and $w \in T'^{-1}(1) \cap U$. It remains to show $\mathfrak{q}(c) = \mathfrak{q}^{(m)}_\lambda(w) + 2[\varepsilon(w) = -1]$ for any c with $c + c^{q^m} = w$.

Step 3: the Witt value of c. Let $N = c\,c^{q^m} \in U$. By transitivity of the trace of Witt vectors and $\varphi([c]) = [c^{q^m}]$,

$$\mathsf{T}_{2mk}(c) = \mathrm{Tr}_{W_2(U)/\mathbb{Z}/4}\big([c] + [c^{q^m}]\big) = \mathrm{Tr}_{W_2(U)/\mathbb{Z}/4}(w, N) = \mathsf{T}_{mk}(w) + 2\,\mathrm{Tr}_{q^m/2}(N),$$

using $[x]+[y]=(x+y,xy)$ in W_2 of a ring of characteristic 2 and $(w,N)=[w]+V[N]$. The power sums $p_j=c^j+c^{jq^m}$ of the two conjugates are, by Newton's identities with $e_1=w$, $e_2=N$: $p_1=w$, $p_3=w^3+Nw$, $p_5=w^5+Nw^3+N^2w$. Hence

$$\begin{aligned}\mathfrak{q}(c)&=\mathsf{T}_{mk}(w)+2\,\mathrm{Tr}_{q^m/2}\big(N+\lambda_5p_5+\lambda_3p_3+\lambda_1p_1\big)\\&=\mathfrak{q}_\lambda^{(m)}(w)+2\,\mathrm{Tr}_{q^m/2}\big(N(1+\lambda_5w^3+\lambda_3w)+\lambda_5N^2w\big)=\mathfrak{q}_\lambda^{(m)}(w)+2\,\mathrm{Tr}_{q^m/2}(N\gamma_w),\end{aligned}$$

with $\gamma_w=1+\lambda_5w^3+\lambda_3w+(\lambda_5w)^{1/2}$ (as $\mathrm{Tr}(\lambda_5N^2w)=\mathrm{Tr}(N(\lambda_5w)^{1/2})$). Now $\gamma_w^2=1+\lambda_5^2w^6+\lambda_3^2w^2+\lambda_5w=B_\lambda(w)$, and from $T'(w)=1$, i.e. $\lambda_5w^4+\lambda_3w^2+w=1+(\lambda_3w)^{1/2}+(\lambda_5w)^{1/4}$,

$$w\gamma_w=w+\lambda_5w^4+\lambda_3w^2+(\lambda_5w^3)^{1/2}=1+(\lambda_3w)^{1/2}+(\lambda_5w)^{1/4}+(\lambda_5w^3)^{1/2},\qquad(w\gamma_w)^2=\gamma_w.$$

So $\gamma_w\in\{0,w^{-2}\}$ ($w\neq0$ since $T'(0)=0$), and $\gamma_w=0$ iff $B_\lambda(w)=0$. If w is a root, $\mathfrak{q}(c)=\mathfrak{q}_\lambda^{(m)}(w)$. If w is a total, $N\gamma_w=(c^2+cw)/w^2=v^2+v$ with $v=c/w$, and $v+v^{q^m}=1$, so $v\notin U$ and the Artin–Schreier polynomial $Y^2+Y+(v^2+v)$ over U has no root in U, i.e. $\mathrm{Tr}_{q^m/2}(v^2+v)=1$; thus $\mathfrak{q}(c)=\mathfrak{q}_\lambda^{(m)}(w)+2$. This is the claim, and (6) follows. The description of $T'^{-1}(1)$ is Part III, Theorem 1.1 and §4. □

Remark 3.4. (1) For $N_0\mid4$ the formula with $m=2$ makes s_4, a sum over q^4 elements, a sum of 4 (classes of order 4) or 16 (classes with $N_0\mid2$) terms $\pm i^{\mathfrak{q}^{(2)}(w)}$ times q^2; for $N_0\in\{3,6\}$, $m=3$ gives s_6 as q^3 times 16 (classes $3\,1^3$, 3^2), 8 ($3\,2\,1$) or 4 (class 6) terms. Table 2 lists the patterns at $q=8$. (2) Equivalently: $s_{2m}=\sum_{c\in A}i^{\mathfrak{q}(c)}$ over the affine space $A=\{c\in\mathbb{F}_{q^{2m}}:\mathrm{Tr}_{q^{2m}/q^m}T'(c)=1\}$, a union of $|T'^{-1}(1)\cap\mathbb{F}_{q^m}|$ cosets of $\mathbb{F}_{q^m}$ on each of which $\mathfrak{q}$ is constant; in the classes of order 4, $|A|=4q^2$ and $s_4\neq0$ iff $\mathfrak{q}$ is constant on A (`part4_relative_reduction.py`, verified $q=8$ all pairs, $q=32$ four pairs per class). (3) Formula (6) does not use the values $s_4\in\{0,\pm4q^2\}$ of Part III; in the classes of order 4 it reproves them (Theorem 3.9).

3.3. **Trace parity on the hexad coset.**

Theorem 3.5 (trace parity)**.** *Let $\lambda_5\neq0$, $m\geq1$, $w\in T_\lambda'^{-1}(1)\cap\mathbb{F}_{q^m}$, and suppose $s_m(\lambda_1')\neq0$ for some $\lambda_1'\in\mathbb{F}_q$ (Lemma 3.2(iv)). Then*

$$\mathrm{Tr}_{q^m/2}(w)\equiv mk+f_m\pmod2,\qquad f_m=\dim_{\mathbb{F}_2}\mathcal{K}_\lambda^{\sigma^m}.$$

In particular all elements of $T_\lambda'^{-1}(1)\cap\mathbb{F}_{q^m}$, roots and totals alike, have the same absolute trace.

Proof. Put $s=s_m(\lambda_1')\neq0$ and $\mathfrak{q}=\mathfrak{q}_{\lambda,\lambda_1'}^{(m)}$. The shift identity with $c=w$, $T'(w)=1$, and Lemma 3.2(ii) give

$$s=i^{\mathfrak{q}(w)}\,s_m(\lambda_1'+1)=i^{\mathfrak{q}(w)}\,\bar s.$$

By Lemma 3.2(i),(iii), $s=\mathrm{Re}+i\,\mathrm{Im}$ with $\mathrm{Re}^2+\mathrm{Im}^2=2^{mk+f_m}$ and $\mathrm{Re},\mathrm{Im}\in\{0,\pm2^j\}$. If $mk+f_m$ is even, a power of 4 is a sum of two squares of this shape only if one of them is 0: s is real or purely imaginary, $\bar s=\pm s$, so $i^{\mathfrak{q}(w)}=\pm1$, $\mathfrak{q}(w)$ is even and $\mathrm{Tr}_{q^m/2}(w)\equiv\mathfrak{q}(w)\equiv0$. If $mk+f_m$ is odd, then $\mathrm{Re}^2=\mathrm{Im}^2=2^{mk+f_m-1}$, so $s/\bar s=(\mathrm{Re}+i\,\mathrm{Im})^2/(\mathrm{Re}^2+\mathrm{Im}^2)=\pm i$, $\mathfrak{q}(w)$ is odd and $\mathrm{Tr}_{q^m/2}(w)=1$. □

Corollary 3.6 (rational roots)**.** *Let k be odd and $r\in\mathbb{F}_q$ a root of B_λ (equivalently of $F_{a,b}$) of class C. Then $\mathrm{Tr}_{q/2}(r)\equiv f_1(C)+1\pmod2$: the rational roots have trace 0 in the classes $2\,1^4$, $3\,2\,1$, $4\,1^2$ and trace 1 in the classes 1^6, $2^2\,1^2$, $3\,1^3$, $5\,1$. The same holds for the rational synthematic totals. For even k the trace is $f_1(C)$ mod 2.*

Proof. $m=1$ in Theorem 3.5, the hypothesis being automatic by Lemma 3.2(iv); f_1 is read off Table 1. □

TABLE 2. The signed terms $(\varepsilon(w), \mathfrak{q}_\lambda^{(m)}(w))$ of the hexad formula at $q = 8$, all $(\lambda_3, \lambda_5, \lambda_1)$ of each class (`part4_hexad_formula.py`); "a:b" means a roots and b totals with the given Witt value. The last column is s_{2m}/q^m.

class	m	roots: values	totals: values	s_{2m}/q^m
$2^2 1^2$	2	2:0, 4:2	6:0, 4:2	-4 (12 triples)
		4:0, 2:2	4:0, 6:2	$+4$ (6)
		6:2	10:2	$+4$ (6)
2^3	2	4:0, 2:2	4:0, 6:2	$+4$ (6)
		6:2	10:2	$+4$ (2)
$4\,1^2$	2	2:0	2:2	$+4$ (24)
		2:0	2:0	0 (24)
$4\,2$	2	2:2	2:0	-4 (24)
		2:0	2:0	0 (24)
$3\,1^3$	3	6:1	10:1	$-4i$ (3)
		2:1, 4:3	6:1, 4:3	$-4i$ (9)
		4:1, 2:3	4:1, 6:3	$+4i$ (9)
		6:3	10:3	$+4i$ (3)
3^2	3	6:1	10:1	$-4i$ (12)
		6:3	10:3	$+4i$ (12)
$3\,2\,1$	3	1:0, 3:2	3:0, 1:2	-4 (18)
		3:0, 1:2	1:0, 3:2	$+4$ (18)
		4:0 or 4:2	4:0 or 4:2	0 (36)
6	3	—	4:1	$-4i$ (48)
		—	4:3	$+4i$ (48)

Remark 3.7. (1) The corollary is verified for every (a, b) and every rational root at $q = 8, 32, 128$ (`part4_trace_parity.py`: e.g. at $q = 128$ the 4032 rational roots of the class $4\,1^2$ all have trace 0 and the 3277 of the class $5\,1$ all have trace 1). It is compatible with, and sharper than, the pair lemma of Part III ($\mathrm{Tr}(r+s) = 0$ for two rational roots), which it explains: two rational roots of one sextic lie in the same class. It is also compatible with $\sum N_1 = (q-1)^2$: rational roots of both traces exist in abundance, distributed over the classes according to the parity of f_1. (2) For $m = 2$ in the classes of order 4, $R_2 = \{0, r_1+r_3, r_2+r_4, \tau_0\}$, $\tau_2(r_1+r_3) = \tau_0 \neq 0$ and $\tau_2(\tau_0) = 0$, so $R_2 \cap \ker \tau_2 = \{0, \tau_0\}$ and the hypothesis holds because $\mathfrak{q}^{(2)}(\tau_0) = 2\,\mathrm{Tr}_{q/2}(\tau_0) = 0$ (pair lemma); the conclusion $\mathrm{Tr}_{q^2/2}(w) = 0$ for the four elements of $T'^{-1}(1) \cap \mathbb{F}_{q^2}$ is visible in Table 2. For $m = 3$ in the classes 3^2 and 6, τ_3 vanishes identically on R_3, so the hypothesis is the nonvanishing of s_3 for *some* (equivalently every) λ_1; see §3.7.

3.4. **The classes of order 4.** Let B_λ have class $4\,1^2$ or $4\,2$, with 4-cycle $(r_1 r_2 r_3 r_4)$, remaining roots r_5, r_6, $\tau_0 = r_5 + r_6 \in \mathbb{F}_q^\times$, $\rho_0 = r_5 r_6 \in \mathbb{F}_q$. The quadratic $u^2 + \tau_0 u + \rho_0$ splits over $\mathbb{F}_q$ iff $\mathrm{Tr}_{q/2}(\rho_0/\tau_0^2) = 0$ iff the class is $4\,1^2$ (substitute $u = \tau_0 y$). Since $r_1+r_2+r_3+r_4 = \tau_0$ (the six roots sum to 0),

$$\mathcal{K}_\lambda^{\sigma^2} = \{0,\ r_1+r_3,\ r_2+r_4,\ \tau_0\}, \qquad T_\lambda'^{-1}(1) \cap \mathbb{F}_{q^2} = \{r_5,\ r_6,\ w_3,\ w_4\},$$
$$w_3 = r_5 + r_1 + r_3, \qquad w_4 = r_5 + r_2 + r_4 = w_3 + \tau_0;$$

w_3, w_4 are the two synthematic totals of the synthemes $\{13|24|56\}$ (they are not roots: $w_3 = r_j$ would force two roots to coincide). In the class $4\,2$, $\sigma(w_3) = r_6 + r_2 + r_4 = w_3$, so $w_3, w_4 \in \mathbb{F}_q$; in the class $4\,1^2$, $\sigma(w_3) = w_4$, so $w_3, w_4 \in \mathbb{F}_{q^2} \setminus \mathbb{F}_q$ are conjugate with $w_3 + w_4 = \tau_0$.

Lemma 3.8 (the roots outside the 4-cycle)**.** *For every* λ_1*:* $\mathfrak{q}_\lambda^{(2)}(\tau_0)=0$, $\mathfrak{q}_\lambda^{(2)}(r_6)=\mathfrak{q}_\lambda^{(2)}(r_5)$, $\mathfrak{q}_\lambda^{(2)}(w_4)=\mathfrak{q}_\lambda^{(2)}(w_3)$*, and all four values* $\mathfrak{q}_\lambda^{(2)}(r_5)$, $\mathfrak{q}_\lambda^{(2)}(r_6)$, $\mathfrak{q}_\lambda^{(2)}(w_3)$, $\mathfrak{q}_\lambda^{(2)}(w_4)$ *are even. Moreover* $\mathfrak{q}_\lambda^{(4)}(r_5)=\mathfrak{q}_\lambda^{(4)}(r_6)=\mathfrak{q}_\lambda^{(4)}(\tau_0)=0$.

Proof. $\tau_0\in\mathbb{F}_q$, so $\mathfrak{q}_\lambda^{(2)}(\tau_0)=\mathsf{T}_{2k}(\tau_0)+2\,\mathrm{Tr}_{q^2/2}(f(\tau_0))=2\,\mathrm{Tr}_{q/2}(\tau_0)+0=0$ by the pair lemma (Part III, Theorem 1.3). For $x\in\{r_5,w_3\}$, $T'(x)=1$ and $\tau_0\in\mathcal{K}_\lambda$ give $\mathfrak{q}^{(2)}(x+\tau_0)=\mathfrak{q}^{(2)}(x)+\mathfrak{q}^{(2)}(\tau_0)+2\,\mathrm{Tr}_{q^2/2}(\tau_0)=\mathfrak{q}^{(2)}(x)$, which is the second and third claim (in the class $4\,2$ also $r_6=\sigma r_5$ and $\mathfrak{q}^{(2)}$ is σ-invariant). Parity: $\mathrm{Tr}_{q^2/2}(r_5)=\mathrm{Tr}_{q/2}(\tau_2(r_5))$ is $\mathrm{Tr}_{q/2}(\tau_0)=0$ or $\mathrm{Tr}_{q/2}(2r_5)=0$, and likewise for w_3 ($\tau_2(w_3)=\tau_0$ or $2w_3$). The last assertion is the multiplicity formula of Part III, Lemma 6.5: $\mathfrak{q}_\lambda^{(4)}(x)=2\mathfrak{q}_\lambda^{(2)}(x)$ for $x\in\mathbb{F}_{q^2}$, and the three values of $\mathfrak{q}^{(2)}$ are even. □

Theorem 3.9 (the classes of order 4)**.** *Let* k *be odd and* B_λ *of class* $4\,1^2$ *or* $4\,2$*. For every* $\lambda_1\in\mathbb{F}_q$*:*

(i) $s_4(\lambda_1)=2q^2\big(i^{\mathfrak{q}_\lambda^{(2)}(r_5)}-i^{\mathfrak{q}_\lambda^{(2)}(w_3)}\big)$*, both exponents being even;*

(ii) *in the class* $4\,1^2$*:* $\mathfrak{q}_\lambda^{(2)}(r_5)=0$ *and* $s_4(\lambda_1)=2q^2\big(1-(-1)^{\frac12\mathfrak{q}_\lambda^{(2)}(w_3)}\big)\in\{0,+4q^2\}$*;*

(iii) *in the class* $4\,2$*:* $\mathfrak{q}_\lambda^{(2)}(w_3)=0$ *and* $s_4(\lambda_1)=2q^2\big((-1)^{\frac12\mathfrak{q}_\lambda^{(2)}(r_5)}-1\big)\in\{0,-4q^2\}$*;*

(iv) *consequently* $\operatorname{sgn}s_4(\lambda_1)=(-1)^{\mathrm{Tr}_{q/2}(\rho_0/\tau_0^2)}$ *whenever* $s_4(\lambda_1)\neq0$*; the set* $\{\lambda_1: s_4(\lambda_1)\neq0\}$ *is the affine hyperplane* $\mathrm{Tr}_{q/2}(\lambda_1\tau_0)=\kappa_0$ *with*

$$\kappa_0=1+\tfrac12\mathfrak{q}^{(2)}_{\lambda,\lambda_1=0}(w_3)\quad(4\,1^2),\qquad \kappa_0=1+\tfrac12\mathfrak{q}^{(2)}_{\lambda,\lambda_1=0}(r_5)\quad(4\,2);$$

and $W'(\lambda_3,\lambda_5)=+2q^7$ *in the class* $4\,1^2$, $-2q^7$ *in the class* $4\,2$.

Proof. (i) Theorem 3.3 with $m=2$, the description of $T'^{-1}(1)\cap\mathbb{F}_{q^2}$ above, and Lemma 3.8. (ii) In the class $4\,1^2$, $r_5\in\mathbb{F}_q$ is a root of class $4\,1^2$ with $f_1=1$, so $\mathrm{Tr}_{q/2}(r_5)=0$ by Corollary 3.6, and $\mathfrak{q}_\lambda^{(2)}(r_5)=\mathsf{T}_{2k}(r_5)+2\,\mathrm{Tr}_{q^2/2}(f_\lambda(r_5))=2\,\mathrm{Tr}_{q/2}(r_5)+0=0$. (iii) In the class $4\,2$, $w_3\in\mathbb{F}_q$ is a rational element of $T'^{-1}(1)$ of class $4\,2$ with $f_1=1$, so $\mathrm{Tr}_{q/2}(w_3)=0$ by Corollary 3.6 (which applies to totals), and $\mathfrak{q}_\lambda^{(2)}(w_3)=2\,\mathrm{Tr}_{q/2}(w_3)=0$. (iv) The sign statement is (ii), (iii) and the splitting criterion. For the hyperplane, $\mathfrak{q}_\lambda^{(2)}(x)=\mathfrak{q}_{\lambda,0}^{(2)}(x)+2\,\mathrm{Tr}_{q^2/2}(\lambda_1x)=\mathfrak{q}_{\lambda,0}^{(2)}(x)+2\,\mathrm{Tr}_{q/2}(\lambda_1\tau_2(x))$ and $\tau_2(w_3)=\tau_0$ in the class $4\,1^2$, $\tau_2(r_5)=\tau_0$ in the class $4\,2$; so $s_4(\lambda_1)\neq0$ iff $\frac12\mathfrak{q}_{\lambda,0}^{(2)}(x)+\mathrm{Tr}_{q/2}(\lambda_1\tau_0)=1$ for $x=w_3$, resp. r_5. Finally $S''=q^4s_4$ (Part III, Theorem 6.12) and the hyperplane has $q/2$ points, so $W'=\frac{q}{2}\cdot q^4\cdot(\pm4q^2)=\pm2q^7$. □

Remark 3.10. (1) Theorem 3.9 proves the order-4 part of Target Theorem 1.2 and reproves Part III, Theorem 6.12 and Corollary 6.13 in these classes without the L-polynomial argument. Numerically it is confirmed, with all λ_1, for all 12 pairs at $q=8$ and for 15 pairs at $q=32$ including pairs with $\kappa_0=1$ (`part4_order4_all.py`, `part4_brown_invariants.py`); the values $W'=\pm2q^7$ on the whole classes are known for $q\le128$ from Part II, and the complete run of `part4_order4_all.py` over the 240 pairs at $q=32$ (about two hours) is kept in the repository data. (2) The two ingredients are of different nature: the hexad formula is an exact identity between Gauss sums over $\mathbb{F}_{q^4}$ and $\mathbb{F}_{q^2}$; the trace parity theorem is an *arithmetic* statement about the rational points of the coset $T'^{-1}(1)$, proved by playing the shift identity against complex conjugation. The sign of the Brown invariant of $\mathfrak{q}_\lambda^{(4)}$ on $\mathbb{F}_{q^4}/\mathcal{K}_\lambda$ is thus $(-1)^{\mathrm{Tr}(\rho_0/\tau_0^2)}$ because the roots r_5,r_6 and the totals w_3,w_4 exchange their fields of definition between the two classes ($r_5\in\mathbb{F}_q$, $w_3\in\mathbb{F}_{q^2}\setminus\mathbb{F}_q$ in $4\,1^2$; $r_5\in\mathbb{F}_{q^2}\setminus\mathbb{F}_q$, $w_3\in\mathbb{F}_q$ in $4\,2$), while the rational one of the two always has trace 0 and the sign ε distinguishes them. (3) The hyperplane parameter

κ_0 of (iv) coincides with Part III's $\kappa_0 = \frac{1}{2}\mathfrak{q}^{(4)}_{\lambda,0}(r_1+r_5)$, both describing the same set; Proposition 3.12 gives it in closed form.

3.5. **The hyperplane parameter and the quartic cofactor.**

Lemma 3.11 (the quartic cofactor)**.** *Write $B_\lambda/\lambda_5^2 = u^6+au^2+bu+b^2$ with $a=\lambda_3^2/\lambda_5^2$, $b=\lambda_5^{-1}$, and let $u^2+\tau_0u+\rho_0$ be the quadratic factor with roots r_5, r_6. Then*

$$\frac{u^6+au^2+bu+b^2}{u^2+\tau_0u+\rho_0} = u^4+\tau_0u^3+(\tau_0^2+\rho_0)u^2+\tau_0^3u+\frac{b^2}{\rho_0},$$

$$\tau_0 b^2+\rho_0 b+\rho_0^2\tau_0^3=0, \qquad a=\frac{b^2}{\rho_0}+\tau_0^4+\rho_0\tau_0^2+\rho_0^2.$$

The power sums of the roots $r_1,\dots,r_4$ of the cofactor are $p_1=\tau_0$, $p_2=\tau_0^2$, $p_3=\tau_0^3+\rho_0\tau_0$, $p_4=\tau_0^4$, $p_5=\tau_0^5+\rho_0^2\tau_0+b^2\tau_0/\rho_0$, and with $\kappa_0=\frac{1}{2}\mathfrak{q}^{(4)}_{\lambda,\lambda_1=0}(r_1+r_5)=\frac{1}{2}\mathfrak{q}^{(4)}_{\lambda,\lambda_1=0}(r_1)$,

(7)

$$\kappa_0 = e_2^{\mathrm{abs}}(\tau_0)+\mathrm{Tr}_{q/2}\big(\tau_0^2+\rho_0+\lambda_5p_5+\lambda_3p_3\big) = e_2^{\mathrm{abs}}(\tau_0)+\mathrm{Tr}_{q/2}\Big(\tau_0^2+\rho_0+\frac{p_5+\sqrt{a}\,p_3}{b}\Big)\in\mathbb{F}_2,$$

where $e_2^{\mathrm{abs}}(\tau_0)=\sum_{0\le i<j<k}\tau_0^{2^i+2^j}$; if $\tau_0=y^2+y$ with $y\in\mathbb{F}_q$ then $e_2^{\mathrm{abs}}(\tau_0)=\mathrm{Tr}_{q/2}(y+y^3)$. The right-hand side is an $\mathbb{F}_2$-valued trace expression in the ground-field data (τ_0,ρ_0,b,a) alone.

Proof. Long division and comparison of coefficients give the cofactor, the relation between b,τ_0,ρ_0 (the quadratic $b^2+\frac{\rho_0}{\tau_0}b+\rho_0^2\tau_0^2=0$ of Part III, Theorem 1.3, multiplied by τ_0) and the expression for a; Newton's identities in characteristic 2 give the p_j. The identity $\mathfrak{q}^{(4)}(r_1+r_5)=\mathfrak{q}^{(4)}(r_1)$: by (2), $\mathfrak{q}(r_1+r_5)=\mathfrak{q}(r_1)+\mathfrak{q}(r_5)+2\,\mathrm{Tr}_{q^4/2}(r_1)$ with $\mathfrak{q}(r_5)=0$ (Lemma 3.8) and $\mathrm{Tr}_{q^4/2}(r_1)=\mathrm{Tr}_{q/2}(\tau_0)=0$. For (7): by transitivity of the Witt trace, $\mathsf{T}_{4k}(r_1)=\mathrm{Tr}_{W_2(\mathbb{F}_q)/\mathbb{Z}/4}\big(\sum_{i\le4}[r_i]\big)=\mathrm{Tr}_{W_2(\mathbb{F}_q)/\mathbb{Z}/4}\big(\tau_0,\ e_2(r_1,\dots,r_4)\big)=\mathsf{T}_k(\tau_0)+2\,\mathrm{Tr}_{q/2}(\tau_0^2+\rho_0)$, and $\mathsf{T}_k(\tau_0)=\mathrm{Tr}_{q/2}(\tau_0)+2e_2^{\mathrm{abs}}(\tau_0)=2e_2^{\mathrm{abs}}(\tau_0)$; the term $2\,\mathrm{Tr}_{q^4/2}(\lambda_5r_1^5+\lambda_3r_1^3)=2\,\mathrm{Tr}_{q/2}(\lambda_5p_5+\lambda_3p_3)$; and $\lambda_5=1/b$, $\lambda_3=\sqrt{a}/b$. The last assertion is $\mathsf{T}_k(y^2+y)=\mathsf{T}_k(y^2)+\mathsf{T}_k(y)+2\,\mathrm{Tr}(y^3)=2\,\mathrm{Tr}(y)+2\,\mathrm{Tr}(y^3)$, from $[y^2+y]=[y^2]+[y]+(0,y^3)$. □

Proposition 3.12 (closed form of κ_0)**.** *Let $z=b\tau_0/\rho_0=\tau_0/(\lambda_5\rho_0)$, so that $z^2+z=\tau_0^4$ by the relation of Lemma 3.11. Then*

$$\kappa_0 = \mathrm{Tr}_{q/2}\Big(z^3+\frac{\rho_0^2}{z}+\frac{(z+1)\tau_0^2}{\rho_0}\Big). \tag{8}$$

Moreover $\kappa_0+\frac{1}{2}\mathfrak{q}^{(2)}_{\lambda,\lambda_1=0}(r_5)=1$ in the class 4 2, in agreement with Theorem 3.9(iv).

Proof. Put $s=\rho_0/\tau_0^2$ and choose $y\in\mathbb{F}_q$ with $y^4=z$; then $y^2+y=(z^2+z)^{1/4}=\tau_0$, and $b=\rho_0y^4/\tau_0$. In the coordinates (y,s) one has $\rho_0=s\tau_0^2$, $b^2/\rho_0=s\tau_0^4t$ with $t:=(y/(y+1))^4=z/(z+1)$ (indeed $z^2/\tau_0^4=z/(z+1)$), and the three summands of (7) become $e_2^{\mathrm{abs}}(\tau_0)=\mathrm{Tr}(y+y^3)$, $\mathrm{Tr}(\tau_0^2(1+s))$ and, after substituting $p_3=\tau_0^3(1+s)$, $p_5=\tau_0^5(1+s^2+st)$, $1/b=1/(s\tau_0z)$, $\sqrt{a}=\tau_0^2(\sqrt{ts}+1+\sqrt{s}+s)$ and $z=\tau_0^2\sqrt{t}$,

$$\begin{aligned}\mathrm{Tr}\Big(\frac{p_5+\sqrt{a}\,p_3}{b}\Big) &= \mathrm{Tr}\Big(\tau_0^2\,\frac{(1+s^2+st)+(1+s)(\sqrt{ts}+1+\sqrt{s}+s)}{s\sqrt{t}}\Big)\\ &= \mathrm{Tr}\Big(\tau_0^2\Big(1+s+\sqrt{t}+\sqrt{s}+\frac{1}{\sqrt{s}}+\frac{1}{\sqrt{st}}+\sqrt{\tfrac{s}{t}}\Big)\Big).\end{aligned}$$

Using $\mathrm{Tr}(x)=\mathrm{Tr}(x^2)=\mathrm{Tr}(x^4)$, $\tau_0^4=t/(t+1)^2$ and $y=t^{1/4}/(t^{1/4}+1)$, every term is a trace of a rational function of (s,t):

$$\kappa_0=\mathrm{Tr}\Big(\frac{t}{t+1}\Big)+\mathrm{Tr}\Big(\frac{t^3}{(t+1)^3}\Big)+\mathrm{Tr}\Big(\frac{t}{(t+1)^2}\Big(1+s+s^2+t+\frac{1}{s}+\frac{1}{st}+\frac{s}{t}\Big)\Big).$$

With $w=1/(t+1)$, i.e. $t=(1+w)/w$, the three traces are $\mathrm{Tr}(1+w)$, $\mathrm{Tr}((1+w)^3)=1+\mathrm{Tr}(w)+\mathrm{Tr}(w^3)$ (k odd, $\mathrm{Tr}(1)=1$) and $\mathrm{Tr}\big((w+w^2)(\cdots)\big)$; expanding the last one term by term, $(w+w^2)t=(1+w)^2$, $(w+w^2)/(st)=w^2/s$, $(w+w^2)s/t=w^2s$, and the twelve traces collapse to

$$\kappa_0=1+\mathrm{Tr}(w^3)+\mathrm{Tr}(ws^2)+\mathrm{Tr}(w/s).$$

Finally $w=1/(t+1)=(y+1)^4=z+1$, so $\mathrm{Tr}(w^3)=\mathrm{Tr}((z+1)^3)=1+\mathrm{Tr}(z^3)$ (as $\mathrm{Tr}(z^2+z)=\mathrm{Tr}(\tau_0^4)=0$), $\mathrm{Tr}(ws^2)=\mathrm{Tr}((z+1)\rho_0^2/\tau_0^4)=\mathrm{Tr}(\rho_0^2(z+1)/(z^2+z))=\mathrm{Tr}(\rho_0^2/z)$ and $\mathrm{Tr}(w/s)=\mathrm{Tr}((z+1)\tau_0^2/\rho_0)$; this is (8). For the last assertion, in the class $4\,2$ the conjugates r_5,r_6 have $\mathsf{T}_{2k}(r_5)=\mathrm{Tr}_{W_2(\mathbb{F}_q)/\mathbb{Z}/4}(\tau_0,\rho_0)=2e_2^{\mathrm{abs}}(\tau_0)+2\,\mathrm{Tr}(\rho_0)$ and power sums $p_3''=\tau_0^3+\rho_0\tau_0$, $p_5''=\tau_0^5+\rho_0\tau_0^3+\rho_0^2\tau_0$, so $\frac12\mathfrak{q}^{(2)}_{\lambda,0}(r_5)=e_2^{\mathrm{abs}}(\tau_0)+\mathrm{Tr}(\rho_0+\lambda_5p_5''+\lambda_3p_3'')$. Adding (7): $\kappa_0+\frac12\mathfrak{q}^{(2)}_{\lambda,0}(r_5)=\mathrm{Tr}(\tau_0^2)+\mathrm{Tr}\big(\lambda_5(\rho_0\tau_0^3+b^2\tau_0/\rho_0)\big)=0+\mathrm{Tr}\big(\tau_0^4/z+z\big)=\mathrm{Tr}(z+1+z)=\mathrm{Tr}(1)=1$. □

TABLE 3. The parameter κ_0 on the classes of order 4: number of pairs (λ_3,λ_5) with each value (`part4_kappa_scan.py`, from (7) and (8), which agree on every pair; at $q=8,32$ also checked against the direct Witt value, `part4_order4_all.py`). "Pure" means $\lambda_3=0$.

	$4\,1^2$ mixed		$4\,1^2$ pure		$4\,2$	
q	$\kappa_0=0$	$\kappa_0=1$	$\kappa_0=0$	$\kappa_0=1$	$\kappa_0=0$	$\kappa_0=1$
8	3	0	3	0	6	0
32	35	70	5	10	70	50
128	903	1050	21	42	1106	910

Remark 3.13. (1) At $q=8$ all twelve pairs have $\kappa_0=0$, and an earlier draft of this paper conjectured $\kappa_0=0$ in general (equivalently $\mathfrak{q}^{(4)}_{\lambda,0}(r_1)=0$, or $s_4(0)\neq0$); Table 3 refutes this at $q=32$, where 130 of the 240 pairs have $\kappa_0=1$. For those pairs $s_4(0)=0$ and the nonzero values sit on the coset $\mathrm{Tr}(\lambda_1\tau_0)=1$; W' is unaffected, as Theorem 3.9 shows. The counts in Table 3 are not interpolated by quadratics with small denominators, so κ_0 is an arithmetic invariant of the pair beyond its class (the splitting of $\mathbb{F}_q(y)$-type Artin–Schreier conditions in (8)). (2) Formula (8) is the complete description of $\lambda_1\mapsto s_4(\lambda_1)$ in the classes of order 4 in terms of $(\tau_0,\rho_0,\lambda_5)$: $s_4(\lambda_1)=(-1)^{\mathrm{Tr}(\rho_0/\tau_0^2)}\,4q^2\,[\mathrm{Tr}(\lambda_1\tau_0)=\kappa_0]$. See Figure 2.

Lemma 3.14 (the locus of the classes of order 4)**.** *Let k be odd. In the coordinates $y\in\mathbb{F}_q\setminus\{0,1\}$, $s\in\mathbb{F}_q^\times$ of the proof of Proposition 3.12 ($\tau_0=y^2+y$, $\rho_0=s\tau_0^2$, $z=y^4$, $b=\rho_0z/\tau_0$, $t=(y/(y+1))^4$), every pair (a,b) whose sextic has a quadratic factor arises exactly once, and the substitution $u=\tau_0(x^{-1}+1)$ turns the quartic cofactor into the depressed quartic*

$$\Phi(x)=x^4+a'x^2+b'x+c',\qquad a'=\frac{1}{1+t},\quad b'=c'=\frac{1}{s(1+t)}.$$

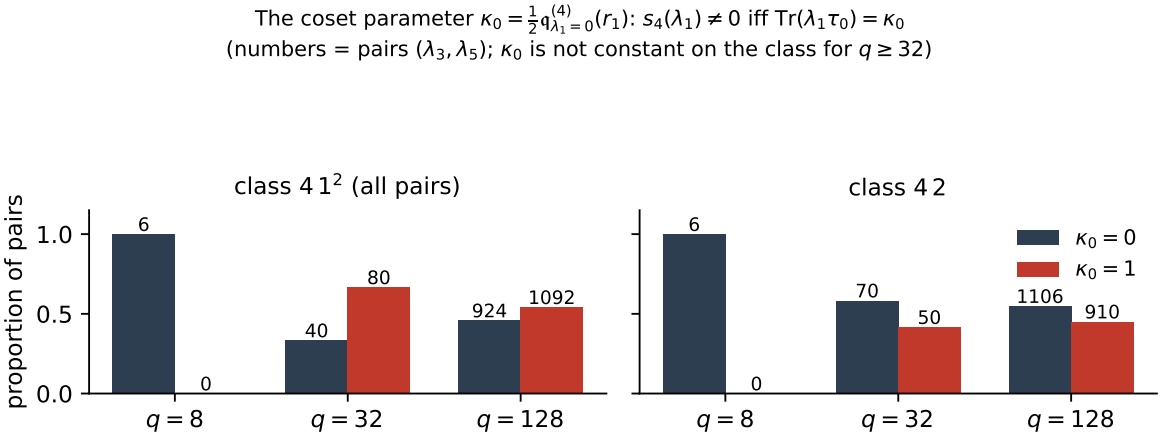


FIGURE 2. Proportion of pairs with $\kappa_0 = 0$ and $\kappa_0 = 1$ in the two classes of order 4 (Table 3).

The pair is of class $4\,1^2$ *or* $4\,2$ *iff* Φ *is irreducible over* $\mathbb{F}_q$*, iff the resolvent cubic* $Y^3+a'Y+b'$ *has exactly one root* $r \in \mathbb{F}_q$ *and*

$$\mathrm{Tr}_{q/2}\Big(\frac{c'}{(a'+r^2)^2}\Big) = 1, \qquad \mathrm{Tr}_{q/2}\Big(\frac{a'}{r^2}\Big) = 0;$$

the class is $4\,1^2$ *iff* $\mathrm{Tr}_{q/2}(s) = 0$.

Proof. The parametrisation is Lemma 3.11 (the two solutions b of $\tau_0 b^2 + \rho_0 b + \rho_0^2\tau_0^3 = 0$ correspond to y and $y+1$). With $u = \tau_0 w$ the cofactor becomes $w^4+w^3+(1+s)w^2+w+st$, with $w = x+1$ this is $x^4 + x^3 + sx^2 + s(1+t)$, and reversing the coefficients gives Φ. For the criterion (Berlekamp [2], Leonard–Williams [20]): if $\alpha_1, \dots, \alpha_4$ are the roots of Φ ($\sum \alpha_i = 0$), the three sums $\gamma = \alpha_i + \alpha_j$ ($= \alpha_l + \alpha_n$ for the complementary pair) are the roots of $Y^3 + a'Y + b'$. A 4-cycle fixes exactly one γ, say $r = \alpha_1 + \alpha_3$; then $\Phi = (x^2+rx+\mu)(x^2+rx+\mu')$ with $\mu = \alpha_1\alpha_3$, $\mu' = \alpha_2\alpha_4$ the roots of $X^2+(a'+r^2)X+c'$, and Φ is irreducible iff $\mu \notin \mathbb{F}_q$ and $x^2+rx+\mu$ has no root in $\mathbb{F}_{q^2}$, i.e. iff $\mathrm{Tr}_{q/2}(c'/(a'+r^2)^2) = 1$ and $\mathrm{Tr}_{q^2/2}(\mu/r^2) = \mathrm{Tr}_{q/2}((a'+r^2)/r^2) = 1$, the latter being $\mathrm{Tr}_{q/2}(a'/r^2) = 1+\mathrm{Tr}_{q/2}(1) = 0$ for odd k. Conversely these conditions exclude the types $(2,2)$, $(1,1,2)$, $(1,3)$, $(1,1,1,1)$ (for which the cubic has three roots, or one root with the opposite trace values). The last assertion is the splitting criterion for $u^2 + \tau_0 u + \rho_0$. All statements are also verified for every (y, s) at $q = 8, 32, 128$ (`part4_kappa_scan.py`). □

Remark 3.15. The lemma describes the locus of the classes of order 4 inside the (y, s)-plane by one cubic and two Artin–Schreier conditions, so the counts $n_{4\,1^2} + n^{\mathrm{pure}}_{4\,1^2} = (q-2)(q-4)/8 + (q-2)/2$ and $n_{4\,2} = q(q-2)/8$ of Target Theorem 1.1 are, in this parametrisation, point counts of an explicit curve of Artin–Schreier type over the cubic resolvent; this is a possible route to those two entries of the table that avoids the moments. The expectation of an earlier draft that the resolvent conditions would force $\kappa_0 = 0$ was wrong: κ_0 and the irreducibility conditions are independent trace conditions.

3.6. **The classes with** $N_0 \mid 2$**.** The sixteen-term version of the hexad formula has a combinatorial structure that decides the sign completely. We first isolate it.

Lemma 3.16 (the sixteen-point configuration)**.** *Let* $m \in \{2, 3\}$*, suppose* $T'^{-1}(1) = r_1 + \mathcal{K}_\lambda \subset \mathbb{F}_{q^m}$ *(all sixteen points rational over* $\mathbb{F}_{q^m}$*), and suppose all sixteen values* $\mathfrak{q}^{(m)}_\lambda(w)$ *have the same parity* $p \in \{0, 1\}$*. Put* $\phi(w) = (\mathfrak{q}^{(m)}_\lambda(w) - p)/2 \in \mathbb{F}_2$ *and* $\phi_i = \phi(r_i)$*. Then*

(i) $\phi(r_i+r_j+r_l)=\phi_i+\phi_j+\phi_l$ *for every total, and* $\sum_{i=1}^{6}\phi_i=0$*;*
(ii) *with* $n_1=\#\{i:\phi_i=1\}$ *(an even number),*

$$\sum_{i=1}^{6}(-1)^{\phi_i}-\sum_{w\ \text{total}}(-1)^{\phi(w)}=\begin{cases}+4, & n_1\in\{2,6\},\\ -4, & n_1\in\{0,4\};\end{cases}$$

(iii) *consequently* $s_{2m}(\lambda)=i^p\,q^m\cdot(\pm4)$ *with the sign of* (ii)*; in particular* $s_{2m}\neq0$.

Proof. (i) For $\kappa\in\mathcal{K}_\lambda$ and $w\in T'^{-1}(1)$, (2) gives $\mathfrak{q}(w+\kappa)=\mathfrak{q}(w)+\mathfrak{q}(\kappa)+2\,\mathrm{Tr}_{q^m/2}(\kappa)$, where $\mathfrak{q}(\kappa)$ is even (κ is a difference of two coset points of equal parity) and $\mathfrak{q}$ is additive on $\mathcal{K}_\lambda$ (Lemma 3.2(i), $\mathcal{K}_\lambda\subset\mathbb{F}_{q^m}$ being the radical). So $\psi(\kappa):=\frac12\mathfrak{q}(\kappa)+\mathrm{Tr}_{q^m/2}(\kappa)$ is an $\mathbb{F}_2$-linear functional on $\mathcal{K}_\lambda$ and $\phi(w+\kappa)=\phi(w)+\psi(\kappa)$. For a duad, writing $\mathfrak{q}(r_i)=p+2\phi_i$ and using $T'(r_i)=1$, $\mathrm{Tr}_{q^m/2}(r_j)\equiv\mathfrak{q}(r_j)\equiv p$:

$$\mathfrak{q}(r_i+r_j)=\mathfrak{q}(r_i)+\mathfrak{q}(r_j)+2\,\mathrm{Tr}_{q^m/2}(r_j)\equiv 2p+2\phi_i+2\phi_j+2p\equiv 2(\phi_i+\phi_j)\pmod 4,$$

so $\psi(r_i+r_j)=\phi_i+\phi_j+\mathrm{Tr}(r_i)+\mathrm{Tr}(r_j)=\phi_i+\phi_j+2p=\phi_i+\phi_j$. Hence $\phi(r_i+r_j+r_l)=\phi_i+\psi(r_j+r_l)=\phi_i+\phi_j+\phi_l$. Computing the same total through the complementary triple gives $\sum_{i=1}^{6}\phi_i=0$; equivalently, $\mathfrak{q}(\sum r_i)=\mathfrak{q}(0)=0$ expanded by (2). (ii) $\sum_i(-1)^{\phi_i}=6-2n_1$. The ten totals correspond to the ten partitions of $\{1,\dots,6\}$ into two triples, and the sign of a total is $(-1)^t$ with t the number of indices with $\phi=1$ in either triple (t and n_1-t have the same parity as n_1 is even). For $n_1=0$: all totals $+$, total $6-10=-4$. For $n_1=2$: the two indices lie in the same triple for 4 partitions ($+$) and are separated for 6 ($-$): $2-(4-6)=+4$. For $n_1=4,6$ replace ϕ by $\phi+1$, which negates both sums. (iii) is Theorem 3.3. □

Theorem 3.17 (the classes with $N_0\mid 2$)**.** *Let k be odd and B_λ of class 1^6, $2\,1^4$, $2^2\,1^2$ or 2^3. For every λ_1 all sixteen values $\mathfrak{q}^{(2)}_\lambda(w)$, $w\in T'^{-1}(1)$, are even; put $\phi_i=\frac12\mathfrak{q}^{(2)}_\lambda(r_i)$. Then $s_4(\lambda_1)=\pm4q^2$ with the sign of Lemma 3.16*(ii)*, where:*

(i) *every rational root has $\phi_i=\mathrm{Tr}_{q/2}(r_i)=f_1(C)+1$, and conjugate roots have equal ϕ_i;*
(ii) *class 1^6: $n_1=6$ and $s_4(\lambda_1)=+4q^2$ for all λ_1;*
(iii) *class 2^3: the number of conjugate pairs with $\phi=1$ is odd, $n_1\in\{2,6\}$, and $s_4(\lambda_1)=+4q^2$ for all λ_1;*
(iv) *class $2\,1^4$ (rational roots $r_1,\dots,r_4$, pair r_5,r_6, $\tau_0=r_5+r_6$): $n_1=2\phi_5$ and $s_4(\lambda_1)=+4q^2$ iff $\frac12\mathfrak{q}^{(2)}_\lambda(r_5)=1$, which holds for exactly $q/2$ values of λ_1 (those with $\mathrm{Tr}_{q/2}(\lambda_1\tau_0)=1+\frac12\mathfrak{q}^{(2)}_{\lambda,0}(r_5)$);*
(v) *class $2^2\,1^2$ (rational roots r_1,r_2, pairs $\{r_3,r_4\}$, $\{r_5,r_6\}$): $n_1=2+2\phi_3+2\phi_5$ and $s_4(\lambda_1)=+4q^2$ iff $\phi_3=\phi_5$, which holds for exactly $q/2$ values of λ_1 (those with $\mathrm{Tr}_{q/2}(\lambda_1(r_1+r_2))=\frac12\mathfrak{q}^{(2)}_{\lambda,0}(r_3)+\frac12\mathfrak{q}^{(2)}_{\lambda,0}(r_5)$).*

In particular $W'=+4q^7$ in the classes 1^6, 2^3 and $W'=0$ in the classes $2\,1^4$, $2^2\,1^2$: Target Theorem 1.2 holds for the six classes with $N_0\mid 4$.

Proof. Parity. Theorem 3.5 with $m=2$, $2k+f_2=2k+4$ even, once its hypothesis is checked: $\mathcal{K}_\lambda\cap\ker\tau_2=\mathcal{K}_\lambda\cap\mathbb{F}_q=\mathcal{K}_\lambda^\sigma$ is spanned by the duads fixed by σ, i.e. $r+\sigma r$ for a conjugate pair and r_i+r_j for two rational roots, and $\mathfrak{q}^{(2)}_{\lambda,0}$ vanishes on them: $\mathfrak{q}^{(2)}(r+\sigma r)=2\mathfrak{q}^{(2)}(r)+2\,\mathrm{Tr}_{q^2/2}(r)\equiv2(\mathfrak{q}^{(2)}(r)+\mathrm{Tr}_{q^2/2}(r))=0$ by Lemma 3.2(ii), and $\mathfrak{q}^{(2)}(r_i+r_j)=2\,\mathrm{Tr}_{q/2}(r_i+r_j)=0$ for rational r_i,r_j by the pair lemma (for $x\in\mathbb{F}_q$, $\mathfrak{q}^{(2)}_\lambda(x)=2\,\mathrm{Tr}_{q/2}(x)$ as in Step 1 of Theorem 3.3). So Lemma 3.16 applies with $p=0$.

(i) For rational r, $\mathfrak{q}^{(2)}_\lambda(r)=2\,\mathrm{Tr}_{q/2}(r)$ for every λ_1, and $\mathrm{Tr}_{q/2}(r)=f_1+1$ by Corollary 3.6; $\mathfrak{q}^{(2)}_\lambda$ is σ-invariant. (ii) $f_1(1^6)=4$. (iii) $f_1(2^3)=2$; the four rational points of the coset are totals (the partitions $\{1,3,5\}|\{2,4,6\}$ etc. with one index from each pair, which

$\sigma = (12)(34)(56)$ maps to their complements), and such a total $w = r_a + r_b + r_c$ has $\phi(w) = \mathrm{Tr}_{q/2}(w) = f_1 + 1 = 1$ by Corollary 3.6 (totals included) and $\phi(w) = \phi_a + \phi_b + \phi_c$ by Lemma 3.16(i), one ϕ from each pair; hence an odd number of pairs has $\phi = 1$. (iv), (v): $f_1 = 3$, resp. 2, give $\phi = 0$, resp. 1, at the rational roots, and the pairs contribute 0 or 2 to n_1; the λ_1-dependence is $\frac{1}{2}\mathfrak{q}^{(2)}_\lambda(r) = \frac{1}{2}\mathfrak{q}^{(2)}_{\lambda,0}(r) + \mathrm{Tr}_{q/2}(\lambda_1\tau_2(r))$ with $\tau_2(r_5) = \tau_0 \neq 0$ in (iv) and $\tau_2(r_3) + \tau_2(r_5) = r_3 + r_4 + r_5 + r_6 = r_1 + r_2 \neq 0$ in (v), so each sign occurs for $q/2$ values of λ_1. The values of $W' = q^4 \sum_{\lambda_1} s_4$ follow, and the statement about Target Theorem 1.2 combines this with Theorem 3.9. □

Remark 3.18. (1) Table 2 is the theorem at $q = 8$: e.g. in the class $2^2 1^2$ the rational roots always have $\phi = 1$ ("2:2" among the roots), and the three patterns are $n_1 = 2, 4, 6$ with signs $+, -, +$; in the class 2^3 the patterns are $n_1 = 2$ and $n_1 = 6$. The rule is verified for every pair and every λ_1 at $q = 8$ and for three pairs per class at $q = 32$ (`part4_sign_rules.py`). (2) The proof explains the "half-half" behaviour of the classes $2\,1^4$ and $2^2 1^2$ and the constancy in the classes 1^6, 2^3 in one stroke: a conjugate pair contributes an even number to n_1 and its ϕ depends on λ_1 through $\mathrm{Tr}(\lambda_1\tau)$, τ the sum of the pair; in the classes 1^6 and 2^3 these τ's sum to zero over the pairs, so $n_1 \bmod 4$ is independent of λ_1, while in the other two classes exactly one τ (resp. one sum of two τ's) is free.

3.7. **The classes with $N_0 \in \{3, 6\}$.** For $N_0 \in \{3, 6\}$ the relevant small field is $\mathbb{F}_{q^3}$ and the relevant hypothesis of Theorem 3.5 is

(9) (H) $s_3(\lambda_1') \neq 0$ for some $\lambda_1' \in \mathbb{F}_q$, i.e. $\frac{1}{2}\mathfrak{q}^{(3)}_{\lambda,0}$ vanishes on $\mathcal{K}_\lambda \cap \mathbb{F}_{q^3} \cap \ker \tau_3$

(Lemma 3.2(iv)).

Lemma 3.19. (H) *holds unconditionally in the classes $3\,1^3$ and 3^2. In the class 6 it is equivalent to the single identity $\mathfrak{q}^{(3)}_{\lambda,0}(r_1 + r_4) = 0$, $r_4 = \sigma^3 r_1$, which does not depend on λ_1 and is equivalent to $\mathrm{Tr}_{q^3/2}(N\gamma_w) = 1$ for $w = r_1 + r_4$, $N = r_1 r_4$, $\gamma_w = 1 + \lambda_5 w^3 + \lambda_3 w + (\lambda_5 w)^{1/2}$; it holds for all pairs of the class at $q = 8$ and $q = 32$.*

Proof. For a duad inside a σ-orbit contained in $\mathbb{F}_{q^3}$, $\mathfrak{q}^{(3)}(r + \sigma r) = 2\mathfrak{q}^{(3)}(r) + 2\,\mathrm{Tr}_{q^3/2}(r) \equiv 0$ by Lemma 3.2(ii). Class $3\,1^3$ ($\sigma = (123)$): τ_3 vanishes on the duads $r_1 + r_2$, $r_2 + r_3$, $r_1 + r_3$ and is $\neq 0$ on the twelve others ($\tau_3(r_4 + r_5) = r_4 + r_5$, $\tau_3(r_1 + r_4) = r_5 + r_6$), so $\mathcal{K}_\lambda \cap \ker \tau_3 = \{0, r_1 + r_2, r_2 + r_3, r_1 + r_3\}$, on which $\mathfrak{q}^{(3)}_0$ vanishes by the first sentence. Class 3^2 ($\sigma = (123)(456)$, $\alpha := r_1 + r_2 + r_3 = r_4 + r_5 + r_6 \in \mathbb{F}_q$): $\tau_3 \equiv 0$ on $\mathcal{K}_\lambda$, and it remains to treat a duad $r_1 + r_4$ across the two cycles. With $\beta = e_2(r_1, r_2, r_3)$, $\beta'' = e_2(r_4, r_5, r_6)$ and $\beta + \beta'' + \alpha^2 = e_2(r_1, \dots, r_6) = 0$,

$$\begin{aligned}\mathsf{T}_{3k}(r_1) + \mathsf{T}_{3k}(r_4) &= \big(\mathsf{T}_k(\alpha) + 2\,\mathrm{Tr}(\beta)\big) + \big(\mathsf{T}_k(\alpha) + 2\,\mathrm{Tr}(\beta + \alpha^2)\big)\\ &= 2\mathsf{T}_k(\alpha) + 2\,\mathrm{Tr}(\alpha) = 4\,\mathrm{Tr}(\alpha) + 4e_2^{\mathrm{abs}}(\alpha) \equiv 0 \pmod 4,\end{aligned}$$

and $2\,\mathrm{Tr}_{q^3/2}\big(f(r_1) + f(r_4)\big) = 2\,\mathrm{Tr}_{q/2}\big(\lambda_5 p_5 + \lambda_3 p_3 + 2\lambda_1\alpha\big)$ with p_3, p_5 the power sums of all six roots, which are $p_3 = 0$ and $p_5 = 5e_5 = b$ (Newton with $e_1 = e_2 = e_3 = 0$); so this is $2\,\mathrm{Tr}(\lambda_5 b) = 2\,\mathrm{Tr}(1) = 2$. By (2), $\mathfrak{q}^{(3)}_0(r_1 + r_4) = \mathfrak{q}^{(3)}_0(r_1) + \mathfrak{q}^{(3)}_0(r_4) + 2\,\mathrm{Tr}_{q^3/2}(r_4) = 2 + 2\,\mathrm{Tr}_{q/2}(\alpha)$. Now α is a synthematic total ($\{123\}|\{456\}$), rational, so $\mathrm{Tr}_{q/2}(\alpha) = f_1(3^2) + 1 = 1$ by Corollary 3.6, and $\mathfrak{q}^{(3)}_0(r_1 + r_4) = 0$. Class 6: $\mathcal{K}_\lambda \cap \mathbb{F}_{q^3} = \mathcal{K}_\lambda^{\sigma^3} = \{0, r_1 + r_4, r_2 + r_5, r_3 + r_6\}$, on which τ_3 vanishes ($\tau_3(r_1 + r_4) = \sum r_i = 0$) and which is permuted by σ, so (H) is the one identity stated. For the reformulation apply Step 3 of the proof of Theorem 3.3 to $c = r_1$, $c + c^{q^3} = w$, which gives $\mathfrak{q}^{(6)}_\lambda(r_1) = \mathfrak{q}^{(3)}_\lambda(w) + 2\,\mathrm{Tr}_{q^3/2}(N\gamma_w)$ (that step used nothing about $T'(w)$), together with $\mathfrak{q}^{(6)}_\lambda(r_1) = \mathsf{T}_{6k}(r_1) +$

$2\,\mathrm{Tr}_{q^6/2}(f(r_1)) = \mathrm{Tr}_{W_2(\mathbb{F}_q)/\mathbb{Z}/4}(e_1,e_2) + 2\,\mathrm{Tr}_{q/2}(\lambda_5 p_5 + \lambda_3 p_3 + \lambda_1 p_1) = 0 + 2\,\mathrm{Tr}(\lambda_5 b) = 2$. The numerical statement is Part III, Table 7 ($q = 8$) and the L-polynomial data of Part III at $q = 32$ (`witt_lpoly_k5.txt`: $s_3 \neq 0$ for all 128 triples of the class), and `part4_sign_rules.py`. □

Theorem 3.20 (the classes with $N_0 \in \{3, 6\}$)**.** *Let k be odd.*

(i) *In the classes $3\,1^3$ and 3^2, for every λ_1: all sixteen values $\mathfrak{q}^{(3)}_\lambda(w)$, $w \in T'^{-1}(1)$, are odd, $s_6(\lambda_1) = \pm 4i\,q^3$ with the sign of Lemma 3.16(ii) applied to $\phi(w) = (\mathfrak{q}^{(3)}_\lambda(w) - 1)/2$, and $S''(\lambda_1) = +4q^6$. Hence $W' = +4q^7$.*

(ii) *In the class 6, assume* (H)*. Then for every λ_1 the four values $\mathfrak{q}^{(3)}_\lambda(w)$ at the points of $T'^{-1}(1) \cap \mathbb{F}_{q^3}$ (four synthematic totals) are odd and equal, $s_6(\lambda_1) = \mp 4i\,q^3$ accordingly, and $S''(\lambda_1) = +4q^6$, $W' = +4q^7$.*

(iii) *In the class $3\,2\,1$, assume* (H)*. Then every $w \in T'^{-1}(1) \cap \mathbb{F}_{q^3}$ (four roots and four totals) has $\mathrm{Tr}_{q^3/2}(w) = 0$, all terms of (6) are ± 1, and $s_6(\lambda_1) \in \{0, \pm 4q^3\}$.*

Proof. (i) By Lemma 3.19, Theorem 3.5 applies with $m = 3$ and $3k + f_3 = 3k + 4$ odd: every coset point in $\mathbb{F}_{q^3}$ (all sixteen) has odd Witt value. Lemma 3.16 with $p = 1$ gives $s_6 = i\,q^3(\pm 4)$, in particular $s_6 \neq 0$ and $\nu_6 = -s_6/4q^3 = \mp i$; by Part III, Theorem 6.14(iii), $S'' = -4q^6\nu_6^2 = +4q^6$. (ii) Trace parity with $3k + f_3 = 3k + 2$ odd gives odd values at the four points $w_0 + \mathcal{K}^{\sigma^3}_\lambda$; by (2) and (H), $\mathfrak{q}^{(3)}(w_0 + \kappa) = \mathfrak{q}^{(3)}(w_0) + \mathfrak{q}^{(3)}(\kappa) + 2\,\mathrm{Tr}_{q^3/2}(\kappa) = \mathfrak{q}^{(3)}(w_0) + 0 + 2\,\mathrm{Tr}_{q/2}(\tau_3\kappa) = \mathfrak{q}^{(3)}(w_0)$ for $\kappa \in \mathcal{K}^{\sigma^3}_\lambda$ (the λ_1-part of $\mathfrak{q}^{(3)}(\kappa)$ is $2\,\mathrm{Tr}_{q/2}(\lambda_1\tau_3\kappa) = 0$). So the four terms of (6) are equal, all with $\varepsilon = -1$: $s_6 = -4q^3\,i^{\mathfrak{q}^{(3)}(w_0)} = \mp 4iq^3$, and $S'' = +4q^6$ as before. (iii) Trace parity with $3k + f_3 = 3k + 3$ even. □

Remark 3.21. (1) Theorem 3.20(i) settles Target Theorem 1.2 for the classes $3\,1^3$ and 3^2, and (ii) for the class 6 modulo the single Witt identity (H), which is verified for all pairs at $q = 8$ and $q = 32$. Together with Theorems 3.9 and 3.17, nine of the eleven classes are done unconditionally and one conditionally; Table 2 is the theorem at $q = 8$ (class 3^2: $n_1 \in \{0, 6\}$ only, because $\psi \equiv 0$ there; class $3\,1^3$: all four values of n_1 occur). (2) The class $3\,2\,1$ remains open: (iii) is consistent with the target ($S'' = \pm 4q^6$ in equal numbers) but does not prove it, because when $s_6(\lambda_1) = 0$, which happens (Table 2), $S''(\lambda_1)$ is not determined by s_6; one needs either s_{12} directly or the L-polynomial from $s_1, \ldots, s_4$ as in Part III. (3) Figure 3 shows the normalised Frobenius eigenvalues of the Witt sheaf at $q = 8$ in all classes; the order-4 classes are rotated against each other by $2\pi/8$, which is the sign $(-1)^{\mathrm{Tr}(\rho_0/\tau_0^2)}$ of Theorem 3.9 seen on the eigenvalues ($\omega_j^4 = -q^2\,\mathrm{sgn}\,s_4$).

4. The constant X

Theorem 4.1. *Let k be odd. Suppose that Target Theorem 1.1 holds, that* (H) *of* (9) *holds in the class 6, and that*

(a) *$S''(\lambda_1) = +4q^6$ for exactly $q/2$ values of λ_1 in the class $3\,2\,1$ (equivalently $W' = 0$ there).*

Then $X = q^6 + q^5$ and the variance formula of Corollary 1.3 holds. Of the eleven sign hypotheses of Part III, Corollary 1.5, only (a) and (H) *remain (Theorems 3.9, 3.17, 3.20).*

Proof. Under (a), (H) and Theorems 3.9, 3.17, 3.20 the values of W' on the classes are those of Part III, Theorem 6.9 ($+4q^7$, 0, 0, $+4q^7$, $+4q^7$, $+4q^7$, 0, $+2q^7$, $-2q^7$, $-q^7$, $+4q^7$ in the order of Table 1), and the bookkeeping of Part III, Corollary 1.5, with the counts of Target Theorem 1.1 gives $q^3X = W'(0,0) + \sum_{\lambda_3 \neq 0} W'(\lambda_3, 0) + \sum_{\lambda_5 \neq 0, \lambda_3 = 0} W' + \sum_{\text{mixed}} W' = q^7 + q^8 + (3q^8 - 2q^7) + (q^9 - 3q^8 + q^7) = q^9 + q^8$. □

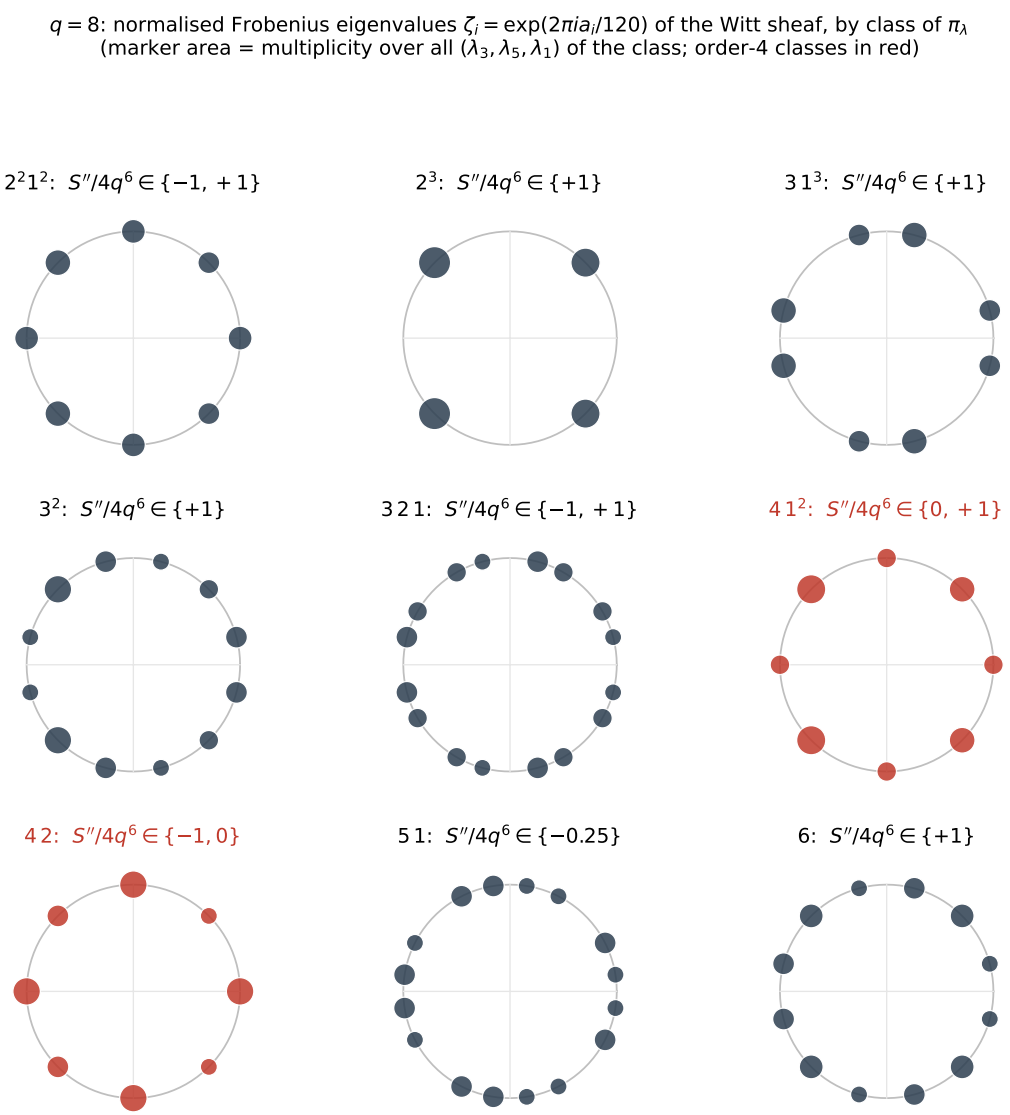


FIGURE 3. Normalised Frobenius eigenvalues $\zeta_j = \omega_j/\sqrt{q}$ of the rank-4 Witt sheaf at $q = 8$, by class of π_λ (from the L-polynomial data of Part III, `witt_lpoly_k3_b120.txt`); marker area is proportional to multiplicity over all $(\lambda_3, \lambda_5, \lambda_1)$ of the class. The two classes of order 4 (red) have $\{\zeta_j\} = \zeta_8^{\pm 1} \cdot \mu_4$ and μ_4 respectively (plus the eigenvalues of the λ_1 with $s_4 = 0$), the mirror image under rotation by $2\pi/8$ predicted by Theorem 3.9.

Remark 4.2 (even k)**.** For even k Part III (§6.11, Table 9) gives $X = -q^6 + q^5$ with class counts depending on k mod 4. The hexad formula and Lemma 3.2 hold verbatim; the trace parity theorem changes by the term mk (even), so rational roots have trace $f_1(C)$ mod 2; in Theorem 3.9 the roles of the two classes are then *exchanged* ($\mathrm{Tr}(r_5) = 1$ in $4\,1^2$, $\mathrm{Tr}(w_3) = 1$ in $4\,2$), giving $s_4 \in \{0, -4q^2\}$ in $4\,1^2$ and $\{0, +4q^2\}$ in $4\,2$, i.e. $W' = -2q^7$ and $+2q^7$, in agreement with the sign $(-1)^{k+1}$ in front of q^6 observed in Part II and with the values $W' = \mp 2q^7$ found at $q = 4$ in Part III, §6.11. At $q = 4$ the four pairs of the classes $4\,1^2$, $4\,2$ indeed have $s_4 \in \{0, -4q^2\}$, resp. $\{0, +4q^2\}$, on the hyperplane $\mathrm{Tr}(\lambda_1\tau_0) = \kappa_0$, with $W' = -2q^7$, resp. $+2q^7$ (`part4_order4_all.py 2`). The even-k form of Corollary 3.6, $\mathrm{Tr}_{q/2}(r) \equiv f_1(C) \pmod 2$, is verified for every rational root at

$q = 4, 16, 64$ (`part4_trace_parity.py`: e.g. all 1024 rational roots of the class $4\,1^2$ at $q = 64$ have trace 1, all 819 of the class $5\,1$ have trace 0).

5. Supersingular abelian surfaces, canonical lifts and the hexad

This section is expository and contains no theorems; it records the interpretation of the structures above that the authors regard as most likely, and the questions it raises (Part III, Open Problems 4 and 6).

5.1. **The hexad as 2-torsion of a lift.** The six roots of B_λ do not depend on λ_1; they are attached to the supersingular curve $C_\lambda : y^2 + y = \lambda_5 x^5 + \lambda_3 x^3$ of genus 2, whose Jacobian J_λ is a supersingular abelian surface (van der Geer–van der Vlugt [12]; for the classification of supersingular abelian surfaces see Oort [23]). The symplectic $\mathbb{F}_2$-space $(\mathcal{K}_\lambda, \omega)$ of Part III has automorphism group $\mathrm{Sp}_4(\mathbb{F}_2) \cong S_6$ [10], the six roots behave as the six Weierstrass points of a genus-two curve (the totals as the ten even theta characteristics, the duads as the fifteen nonzero 2-torsion points), and the Frobenius acts through S_6 as π_λ. In characteristic 2 the Jacobian of a supersingular curve has no étale 2-torsion, so $(\mathcal{K}_\lambda, \omega)$ cannot be $J_\lambda[2](\bar{\mathbb{F}}_q)$. It can be the 2-torsion of a *lift*: by the theory of Serre and Tate [22, 15] every ordinary abelian variety has a canonical lift to $W(\mathbb{F}_q)$; for supersingular ones there are the quasi-canonical lifts of Gross [13] and, more generally, lifts to finite extensions of $W(\mathbb{F}_q)$ whose generic fibre is an abelian surface $\tilde{J}$ with $\tilde{J}[2] \cong (\mathbb{Z}/2)^4$ carrying the Weil pairing and a Galois action through $\mathrm{Sp}_4(\mathbb{F}_2)$. The Witt vectors of length two, which are the natural home of the forms $\mathfrak{q}_\lambda^{(m)}$ (they are $\mathbb{Z}/4$-valued because $W_2(\mathbb{F}_2) = \mathbb{Z}/4$), are the mod-4 reduction of this lifting ring, and the Dieudonné crystal of J_λ reduced modulo 2 is where $\mathcal{K}_\lambda$ should live. The hexad formula (Theorem 3.3) fits this picture: a Gauss sum over $\mathbb{F}_{q^{2m}}$ is computed by a sum over the $\mathbb{F}_{q^m}$-rational 2-torsion-like points (roots and totals) of the lift, with the sign ε distinguishing the "odd" from the "even" points, exactly as the theta constants of a genus-two curve distinguish the six odd from the ten even characteristics; the trace parity theorem is then a statement about the field of definition of these points.

5.2. **The role of μ_5 and the dependence on k mod 4.** For even k the class counts of Part III, Table 9, depend on k mod 4 through the classes 1^6, $2^2 1^2$, $5\,1$ and the pure column, while for odd k they do not. The mechanism is the splitting of the fifth roots of unity: $\mu_5 \subset \mathbb{F}_q$ iff $4 \mid k$, $\mathbb{F}_q(\mu_5) = \mathbb{F}_{q^2}$ for $k \equiv 2 \bmod 4$, and $\mathbb{F}_q(\mu_5) = \mathbb{F}_{q^4}$ for odd k regardless of k mod 4. The class $5\,1$ corresponds to a quintic factor of B_λ whose splitting field is $\mathbb{F}_{q^5}$ and whose Galois group, a 5-cycle, is the image of a Kummer-type extension generated by a fifth root once μ_5 is adjoined; the 5-cycles form one class in S_6 but two in A_6, and which of the two occurs is governed by $\mathrm{Gal}(\mathbb{F}_q(\mu_5)/\mathbb{F}_q)$. Through the moment identities (Proposition 2.8) the branching of $n_{5\,1}$ along k mod 4 propagates to 1^6 and $2^2 1^2$. In the picture of the previous paragraph, μ_5 enters through the automorphism $x \mapsto \zeta_5 x$ of the pure curves $y^2 + y = \lambda_5 x^5$ (complex multiplication by $\mathbb{Z}[\zeta_5]$ on the lift), and the decomposition group of 5 in $\mathbb{Q}(\zeta_5)/\mathbb{Q}$ at the prime 2 (which has order 4) is what makes $m = 5$ special in the moment system: the row $m = 5$ of Proposition 2.8 is the only one whose Gauss-sum total $\Sigma_5 = q^3(q^2 + 1)$ involves $q^2 + 1 = (q + i)(q - i)$, the norm form of the splitting of 2 in $\mathbb{Q}(i) \subset \mathbb{Q}(\zeta_{20})$, and the only one in which $f^{(5)}$ separates the class $5\,1$ from all others. A proof of the even-k counts would identify the relevant Kummer extension explicitly; the odd-k counts, where $\mathbb{F}_q(\mu_5)$ is always $\mathbb{F}_{q^4}$, are the case without branching.

Computations

All scripts are in the directory `code/` of the repository and depend only on NumPy, SymPy and Numba (for the field tables); their outputs are in `data/`, the figures are produced by `paper/make_figures4.py`.

- `moment3_check.py`: Proposition 2.1 and Conjecture 2.4 for $q = 8, 32, 128$.
- `moment3_structure.py`: Proposition 2.2.
- `part4_moment_system.py`: Proposition 2.8 and the interpolation checks.
- `part4_hexad_formula.py`: Theorem 3.3 for every λ at $q = 8$ ($m = 2, 3$) and for three pairs per class with $N_0 \mid 4$ at $q = 32$ ($m = 2$); Table 2.
- `part4_trace_parity.py`: Corollary 3.6 for every (a, b) at $q = 8, 32, 128$ (and $q = 4, 16, 64$ for even k).
- `part4_sign_rules.py`: Lemma 3.16 and Theorems 3.17, 3.20 (ϕ-values with the Frobenius orbit structure, n_1, predicted and actual s_{2m}, hypothesis (H)) for every pair and every λ_1 at $q = 8$ and three pairs per class at $q = 32$.
- `part4_order4_all.py`: for every pair of the classes of order 4 at $q = 8$ and $q = 32$, the direct Witt values, κ_0 by (7) and (8), and $s_4(\lambda_1)$ for all λ_1 (sign, hyperplane, W').
- `part4_kappa_scan.py`: Table 3 and Lemma 3.14; `part4_relative_reduction.py`: the coset form of the hexad formula; `part4_brown_invariants.py`: the earlier sampling version.

Acknowledgements

The computations were carried out with NumPy and SymPy. The exploration that led to the hexad formula and to the trace parity theorem was carried out in an interactive session with Claude (Anthropic), which also assisted with the drafting of the manuscript; all proofs were checked by hand and all numerical statements by the scripts above.

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